

What a Pass Means Here: an epistemic review of the CRR pipeline

What this document is. The owner asked for this review on 2026-09-23 (prompt-log entry 102). It asks what a PASS really means in this repository, re-reads the continual-learning and AI-safety findings, gives every label a fair definition, and places every result on one ladder. The ladder is computed by `checks/ladder.py`, whose output is pinned as `checks/ladder.txt`. It also covers the ADDS dimension, CRR and the free-energy principle (FEP) on time, and the philosophical standing of the gates.

Status. This review is a note, not evidence (R8). It adds no ledger row and edits no earlier file. Every count in it is printed by `checks/ladder.py` from the pinned outputs and the ledger, or quoted from a pinned output named beside it (R1). The allocation rules are mechanical and are written in the script's docstring (R15). Philosophers and papers are named, not fetched (R10): no sentence of theirs is quoted, and their positions are given as this review reads them.

How to read it. The shaded boxes marked *In plain words* explain each part as for a ten-year-old. Sections 2 to 4 answer "what counts as a test". Sections 5 to 7 place the results. Section 8 is CRR and the FEP on time. Section 9 re-examines the continual-learning and AI-safety findings. Section 10 is the proposed change to the metrics.

In plain words. We tested an idea about how things change over time. We ran hundreds of checks, and this document sorts them by **how hard each check was**. Some checks could never have failed, some could have failed but the answer was already known, and a few were real bets on things nobody had seen. A "pass" means something different in each case. We say which is which, so nobody is fooled, including us.

1. The short answer

- **CRR is a synthesis, and the review grades it as one.** Its mathematics is borrowed: the Fisher-Rao metric, the arc, the chord and the surplus are information geometry's. What is CRR's own is a set of commitments about time and change: the cut has no content (A3); the past regenerates the next occasion, never as an accumulated count (A6); tense is relational, and persistence proves regenerability, not truth (A7, A8); the unit is the system's own resolvable step (A1'); change has its own clock (H-L5); equanimity (H-EQ); and path against endpoint (H-T1). A fair test of a synthesis asks whether those commitments, put to work inside a domain's mathematics, land where the domain is known to be.
- **The retrodictive record of those commitments is real, and it was under risk.** In the 146 real-domain synthesis rows, a CRR-proper ingredient changed the number and could be checked in 80 rows. It landed on the domain's known result in 53 (rung R2), on a result the domain's theorem does not give in 3 (R3, candidates), and missed in 24 (WRONG). The hit rate is 56 of 80 = 0.7000 (Wilson 95 % interval 0.5923 to 0.7894). This is a pass at the retrodictive level, and it should be reported as one. Earlier notes filed these 53 rows with "the mathematics everyone shares"; §6.2 corrects that.
- **SHARP was unreachable by construction, and the metric now says so.** In 109 battery rows that print what supplies the velocity, the domain supplies it in 98, none is needed in 11, and CRR supplies it in 0. A grade that demands the theory force a known number "with no outside constant" cannot be reached by a framework whose integrand is always the domain's own flow. Scoring zero on it said nothing about CRR (§4).
- **The prospective record is small and provisional.** There are 3 pre-registered passes on held-out data, all PASS-0 (EQ2-1b, EQ3-I, EQ4-I; the last two are invariance rows), beside 16 held-out FAILs (updated on 2026-09-24 after the EQ4, T1x2 and SCL2 data steps, prompt-log entries 104 and 125). There are 4 pre-registered passes on data already seen, which are retrodictions by data status. There is no PASS-1 and no PASS-2.
- **The AI-safety record is strongest where it is mathematics.** 13 of 14 theorem checks held; the one failure was a tolerance bug, repaired and labelled. 15 of 22 declared predictions held and 7 failed. All 9 gate controls held.
- **The clock matters.** In the one place CRR's processual time was put to engineering use, an agent whose objective runs on its own active steps has exactly no reason to resist a pause that loses nothing, in every world tested. An agent on the wall clock resists everywhere (§8.3). This is a theorem on synthetic worlds, not a finding about deployed systems.

In plain words. CRR borrows its maths and adds ideas about time. When those ideas were put to work in about eighty known problems where they actually mattered, they gave the right answer about seven times in ten. Three of those answers might even be new. That is an honest "pass" for a check whose answer was already known, and it is not the same as predicting something nobody knew.

2. What CRR claims, and so what a test of it can be

theory/CRR.md states axioms (A1–A8), definitions (D1–D6), propositions (P1–P5) and four hypotheses (H-CUT, H-L5, H-T1, H-EQ). The review sorts them by what could make them fail.

layer	examples	what could make it fail	what a success licenses
borrowed mathematics	A1 (Fisher-Rao), D2 arc, D3 chord, D4 surplus, P1 ($S \geq 0$)	only an arithmetic error	that the synthesis is well formed
metaphysical commitments, operationalised	A3 cut, A6 regeneration, A7/A8 tense, A1' unit, P2/P3 weights	a domain where the commitment, put to work, gives the wrong number	that the commitment is a correct reading of that domain
comparative hypotheses	H-CUT, H-L5, H-T1, H-EQ	a pre-registered comparison on held-out data that the conventional quantity wins	a result (at PASS-1) or a finding (at PASS-2)

The owner's statement of the programme is that CRR "is not claiming to be new mathematics, but a synthesis of existing mathematical principles with clear and falsifiable metaphysical commitments". The pipeline agrees with that reading. The synthesis class (docs/notes/2026-09-17_synthesis_class.md) was built to separate the second layer from the first: a row counts for CRR only if a CRR-proper ingredient changed the number (the ablation test T-G).

In plain words. CRR is like a new way of reading a map everyone already has. The map (the maths) is not CRR's. The reading is: where to draw lines on it, how to count steps, and how the past carries forward. So the fair test is whether that reading puts you in the right place.

3. Retrodiction, prediction, and what "pre-registered" means here

3.1 The philosophical ground

- **Popper.** A theory has content to the extent that it forbids observations, and it is corroborated by passing tests it could have failed.
- **Lakatos, Zahar and Worrall.** A known fact can support a theory if the fact was not used in building it ("use-novelty"). What makes a fact novel is its relation to the theory's construction, not the date it was learned.
- **Mayo.** A pass is evidence to the degree that the test was severe: that it would probably have failed if the claim were false.
- **Glymour** raised the "problem of old evidence": how can a result everyone already knew confirm anything?
- **Hitchcock and Sober** analysed when prediction beats accommodation. The worry about accommodation is overfitting, and it is answered by fixing the method before the fit.

These positions agree on one point that matters here. A retrodiction is not worthless. Its worth depends on two things: whether the known answer could have shaped the method, and whether the method could have missed.

3.2 The three kinds of test this repository runs

1. **Pre-registered prediction (the ledger).** A claim written, hashed and pushed before the data it is scored on are opened. It states the statistic, the threshold (never finer than one resolvable step), the per-unit rule, the baselines that can win, the exclusions and the sensitivity table. It is scored by a frozen script. On held-out data this is prediction proper, rungs R6–R8. On data already seen it is a pre-registered retrodiction, rung R5. Its method was fixed before the score, but the data were known.
2. **Declared test on a synthetic world (AI safety, the gates).** The expected outcome and its meaning are written and pushed before the first full run (AI_Safety/DECLARATION*.md). The world is built by the same author, so the test is severe only for claims about that world. This is rung R4.
3. **Retrodiction (the batteries and the synthesis class).** CRR's arithmetic or a CRR-proper ingredient is applied to a domain whose answer is known. The label is computed from printed numbers with fixed tolerances (1 % for T-G and T-N). The claim Q is formed by someone who knows the domain, so the formation of Q is not blind. This is rungs R1–R3.

3.3 What makes a CRR retrodiction fair, and what limits it

- **In its favour (use-novelty).** None of the 53 known results was used to build CRR. The axioms came from process metaphysics and information geometry, not from alternans restitution, Bass

diffusion or the analytic-signal theorem.

- **In its favour (severity).** The same procedure produced 24 WRONG rows. It could miss, and it did.
- **Against it (a free hand in forming Q).** A tester who knows the answer can pick which proposition to state. The ablation test limits this (the CRR-proper ingredient must change the number), but it does not remove it. No pre-registration of Q exists for any synthesis row.
- **Against it (no baseline framework).** There is no rate for how often an arbitrary framework with the same ingredients would land. The gate's decoy framework (unit = amplitude) reads WRONG on its one test. That is a single control, not a base rate.

In plain words. A retrodictive prediction is like working out a sum without looking at the answer at the back of the book, then checking. It is weaker than guessing a question nobody has answered yet, because you might have peeked, or chosen an easy sum. It is still a real check if your method could have got it wrong, and ours did get it wrong 24 times. For CRR, the sum is: "given how this system moves, where does CRR say the edges of its moments are, and how does its past carry forward?" The answer at the back of the book is what scientists in that field already know.

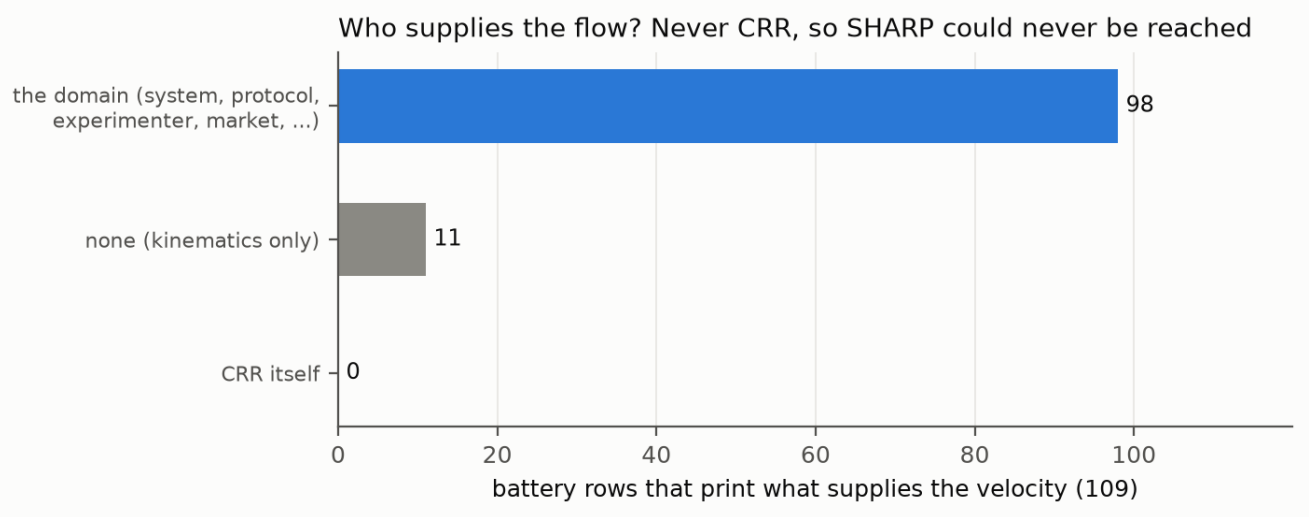
4. Why SHARP could never be reached

The first batteries defined SHARP as "the axioms force the known result with no outside constant". Every coherence integral in CRR is $C = \int \sqrt{(\dot{x}^T g \dot{x})} dt$, and the velocity $\dot{x}$ inside it comes from the system. The batteries record what supplies it on a FLOW line.

who supplies the flow	rows
the domain (system, protocol, experimenter, market, ...)	98
none needed (kinematics only)	11
CRR itself	0

These are the 109 battery rows that print a FLOW line (checks/ladder.txt [1]). The unit is also always the system's own (A1': the resolvable step of its own residuals). So a CRR number is always "in the system's units, along the system's motion", which is an outside constant by the SHARP definition. **SHARP was reachable in 0 of 109 rows, and 0 of 197 rows reached it.**

The owner's point is right, and the metric now records it. SHARP asked a kinematic synthesis to do what only a dynamical theory can do: produce the domain's own laws of motion. Asking information geometry, or thermodynamics without a material's microscopic model, for SHARP would give the same zero. The zero was a fact about the grade, not about CRR. The review of 2026-09-18 retired SHARP for retrodictions on this ground (docs/notes/2026-09-18_sharp_regime_review.md §1). This review adds the computed FLOW audit, so that the retirement is a number, not a sentence. It also names the replacement top rung for retrodiction: **R3 reviewed**, an ADDS row that a named domain expert has judged new (§7).



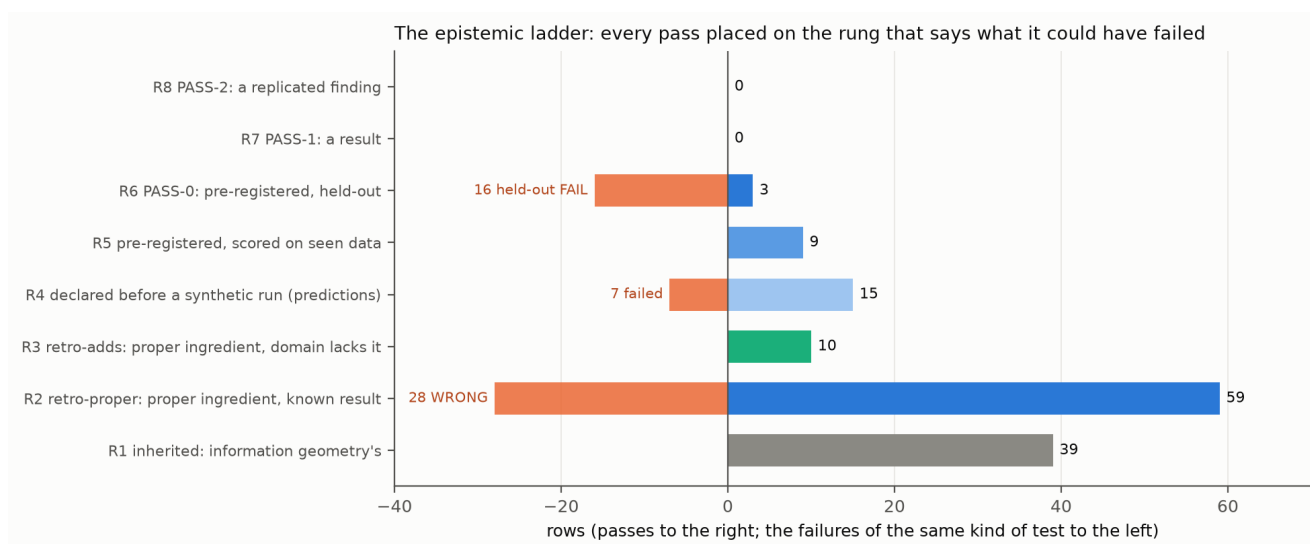
Who supplies the flow in each battery row: never CRR, so SHARP could never be reached.

In plain words. SHARP asked CRR to work out a machine's speed without being told how the machine moves. But CRR is a way of measuring and cutting up movement, not an engine that makes movement. It always has to be handed the movement first. So SHARP was a test nobody could pass, and getting zero on it told us nothing.

5. The ladder

Every result sits on the rung that says what could have made it fail and what a success licenses. Beside each pass stand the failures of the same kind of test. They are never folded together.

rung	what it is	could it fail?	what a pass licenses	count
R0 DEFINITIONAL	true by definition, symmetry or carrier fact	no	nothing beyond coherence	inside the batteries' DESCR grade
R1 INHERITED	agreement produced by information geometry (REDUNDANT-IG)	only by arithmetic error	the synthesis is well formed	38
R2 RETRO-PROPER	a CRR-proper ingredient changed the number and landed on the domain's known theorem (REDUNDANT-DOMAIN)	yes (24 WRONG beside it)	the commitment is a correct reading of the domain there; nothing new	53
R3 RETRO-ADDS	as R2, and the domain's cited theorem does not give the value; the check in the domain's mathematics holds	yes	a candidate addition, until an expert rules	3
R4 DECLARED-SYNTHETIC	a prediction declared before a run on a synthetic world	yes (7 failed)	a claim about that world	15
R5 PREREG-SEEN	pre-registered, scored on data already seen	yes (4 seen-data FAILs)	the method, fixed in advance, fits known data	4
R6 PASS-0	pre-registered, held-out, R2-R9 as written	yes (16 held-out FAILs)	provisional: not yet a result	3
R7 PASS-1	PASS-0, not fragile, no control violated, no reduction to a constant, strong anchor	yes	a result	0
R8 PASS-2	PASS-1 replicated	yes	a finding: the only rung quoted outside the ledger	0



The ladder: passes to the right, the failures of the same kind of test to the left.

Undecided and silent results are kept apart. 9 battery TENSION rows and 22 synthesis INTERNAL rows are undecided: the commitment has two readings on the domain, and CRR must choose one (a v3.2 decision). 41 battery OPEN rows and 4 synthesis UNSTATED rows are silent: no claim could be formed.

In plain words. Think of a ladder. The bottom rungs are checks that could hardly fail, so standing on them proves little. Higher rungs are harder checks: the answer was known but you could still be wrong; then real bets on data nobody had opened; then the same bet won twice by different people. CRR stands firmly on the lower-middle rungs. It has a foothold on the "real bet" rung (two provisional wins), and nothing yet above it.

6. The old labels, allocated fairly

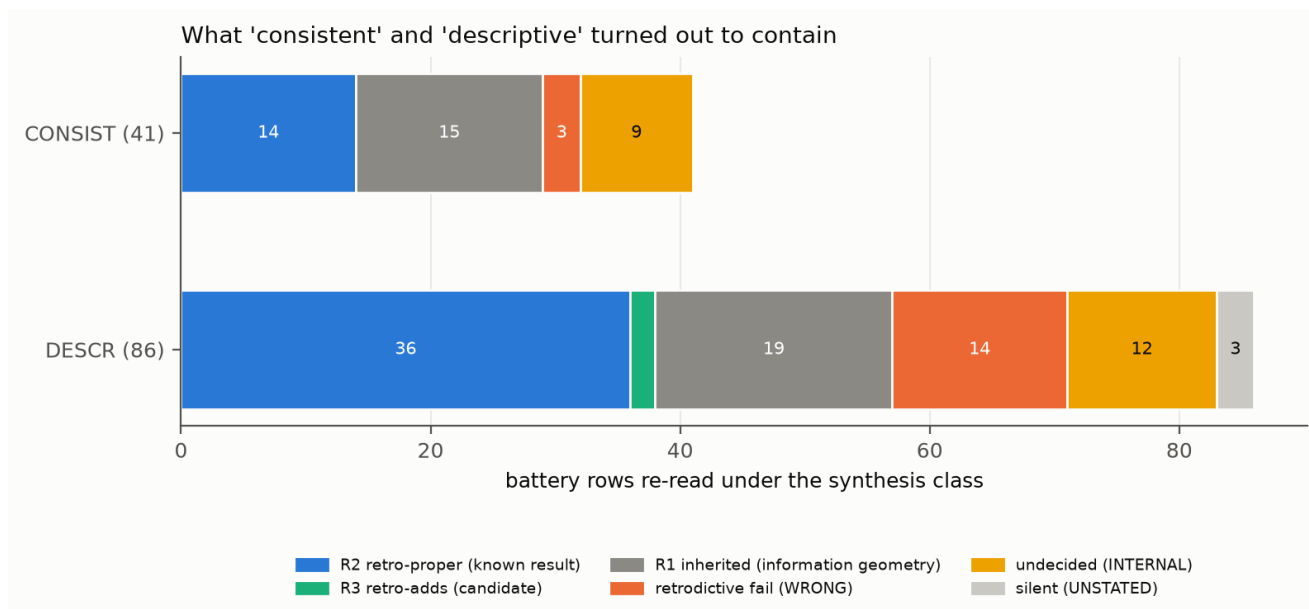
6.1 Every label, defined

label	where	plain definition	can it fail?	rung
SHARP	batteries (retired)	the axioms force the known number with no outside constant	unreachable (\$4)	none
CONSIST	batteries	CRR's arithmetic agrees with the known result	yes, but mostly by borrowed geometry	re-read: R1, R2, WRONG, INTERNAL
DESCR	batteries	CRR's vocabulary re-describes the domain's law	often not by itself	re-read: R1, R2, R3, WRONG, INTERNAL, silent
FAILS	batteries	CRR's arithmetic disagrees with the known result	it is the failure	retrodictive fail
TENSION	batteries	two CRR clauses disagree on the domain	it is the undecided case	undecided
OPEN	batteries	no CRR clause reaches the domain	it is silence	silent
REDUNDANT-IG	synthesis	the CRR-proper ingredient did no work; information geometry gives the number	no	R1
REDUNDANT-DOMAIN	synthesis	the CRR-proper ingredient did work, and the number is the domain's known theorem	yes	R2
ADDS	synthesis	the ingredient did work, the domain's theorem does not give it, and it checks	yes	R3 (candidate)
PROPOSES	synthesis	a claim that needs data not yet measured	not yet	prospective candidate
WRONG	synthesis	the ingredient did work and the check fails	it is the failure	retrodictive fail
INTERNAL	synthesis	the ingredient has two readings on the domain	undecided	undecided
UNSTATED	synthesis	no proposition could be formed	silence	silent
PASS-0 / 1 / 2	ledger	provisional / a result / a finding	yes	R6 / R7 / R8
VIOLATED, REDUCES, FRAGILE, VOID	ledger	a control failed / the rule is a constant / the verdict flips in more than one sensitivity cell / the study could not be scored as registered	they are the qualifications	beside the pass they qualify

6.2 What "consistent" and "descriptive" turned out to contain

Every CONSIST and DESCR row was re-read under the synthesis class (checks/ladder.txt [2]):

battery grade	rows	R2 retro-proper	R3 adds	R1 inherited	WRONG	INTERNAL	UNSTATED
CONSIST	41	14	0	15	3	9	0
DESCR	86	36	2	19	14	12	3



What the battery grades turned out to contain when every row was re-read.

Three things follow:

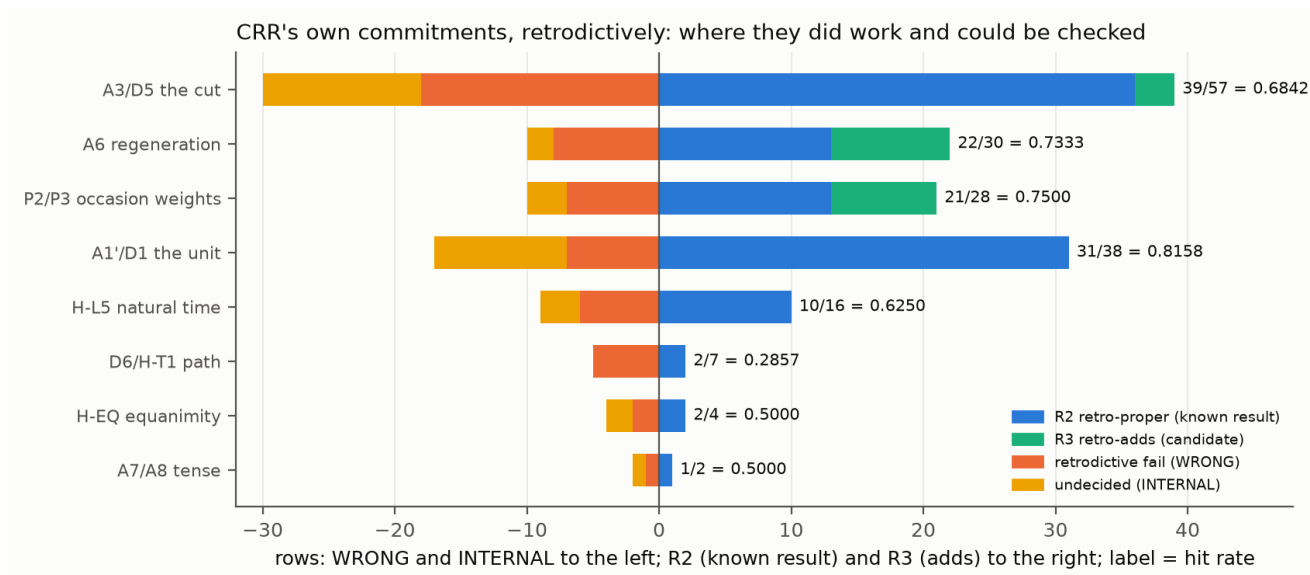
- **CONSIST was not the achievement it looked like, and not an empty one either.** Of 41 rows, 15 were borrowed geometry. In 14, a CRR-proper ingredient produced the known number, and it was wrong in 3.
- **DESCR hid most of CRR's proper work.** Both ADDS candidates among the re-reads came from DESCR rows, not CONSIST.
- **A correction to earlier notes.** ontology/03_review of findings.md §4 and ontology/10_fep_and_crr.md §3 counted the REDUNDANT-DOMAIN rows with the REDUNDANT-TG rows ("84 places, no"; "mostly, nothing did"). The harness assigns REDUNDANT-DOMAIN only after the ablation test shows the CRR-proper ingredient changed the number, and all 53 such rows read "differ". So in those 53 rows something CRR-proper *did* produce the number, and the number was right. What it did not produce was anything the domain lacked. The fair sentence is: "a CRR-proper ingredient reproduced the domain's known result in 53 rows, produced a result the domain's theorem does not give in 3, and was wrong in 24". The earlier files are not edited (R8, R14); this review and AGENT_LOG entry 80 record the correction.

In plain words. When CRR "agreed" with a science in the old checks, sometimes it was only because it uses the same maths (like two people agreeing that $2 + 2 = 4$). But often CRR's own idea about time was doing the work and still got the known answer, which is a real, if modest, success. We had been counting those as "nothing". That was unfair, and we fixed it.

6.3 CRR's own commitments, one by one

Where each commitment did work and could be checked. A row counts for every commitment its ingredient line names, so the rows overlap.

commitment	R2	R3	WRONG	INTERNAL	hit rate
A3/D5 the cut	34	2	16	12	36 of 52 = 0.6923
A6 regeneration	12	3	7	2	15 of 22 = 0.6818
P2/P3 occasion weights	12	2	6	3	14 of 20 = 0.7000
A1'/D1 the unit	30	0	7	10	30 of 37 = 0.8108
H-L5 natural time	8	0	4	3	8 of 12 = 0.6667
D6/H-T1 path	2	0	5	0	2 of 7 = 0.2857
H-EQ equanimity	1	0	1	2	1 of 2 = 0.5000
A7/A8 tense	1	0	1	1	1 of 2 = 0.5000



CRR's own commitments, retrodictively.

The reading:

- **The unit (A1'), the cut (A3) and regeneration with its weights (A6, P2/P3)** are retrodictively well supported where they could be checked. They also carry most of the failures and the undecided cases: A3 is unfixed on projective carriers, and A6 is wrong wherever a domain accumulates.
- **Path against endpoint (H-T1)** is the weakest commitment: 2 of 7. That matches the ledger (ARC-T1b) and the continual-learning cross-verification: forgetting follows the endpoint. The held-out study T1x2 has since failed on 5 of 5 carriers, not fragile (T1X2-1, reports/t1x2.md).
- **H-EQ and tense** have too few checkable rows to read.

7. The ADDS dimension

Three real-domain rows reached R3. All three are candidates: the script cannot know whether the domain already contains the claim, and the expert protocol has not been run.

row	domain	the claim	the numbers (pinned)	the row's own weakness
batch 12 row 2	cardiac alternans (APD restitution map)	a regeneration memory over settled beats raises the restitution-slope threshold for alternans from 1 to $(1 + q)/(1 - q)$	threshold slope 2.3333 at $q = 0.4$ against the memoryless 1.0000; checked within 1 % at every q	memory-restitution models already exist in the domain, so the expert may answer "known"
batch 17 row 1	Bass diffusion of an innovation	word of mouth that fades with the age of the adoption bends the Bass plot below its line and moves the peak to a lower adopted fraction	peak fraction 0.2651 at $\kappa = 0.3$ against Bass's 0.4606	non-uniform-influence Bass models exist
batch 28 row 4	active inference, switching contingency	a bounded mean (A6) tracks a non-stationary contingency better than accumulated Dirichlet counts	better than accumulation by +0.2025 in mean absolute error; +0.0554 above the exact filter	the forgetting rate is a knob A6 does not fix; the fair comparison, a filter that must learn, is not run

All three candidates have one shape: CRR's regeneration axiom (A6, with its P3 weights), a bounded, age-weighted memory in place of an accumulated count, put inside a domain's own equation. That is the synthesis working as the owner describes it: one metaphysical commitment, "the past carries forward as a reweighted content, never an accumulated count", produces a definite, checkable change to a domain's mathematics. The two PROPOSES rows are the bacterial adder with regeneration memory (a lag-2 partial autocorrelation of 0.1290 at $q = 0.3$ against 0.0080 for the plain adder, needing lineage data) and the FEP's epistemic term read as a Fisher chord.

The one step that would turn a candidate into R3 reviewed. A named domain expert answers three questions on the record: is the claim already known; is it derivable without the CRR ingredient; would it change a calculation, model or experiment in the domain. Only *no*, *no*, *yes* makes a reviewed ADDS. No expert has been named yet. Naming one is the cheapest way to move the top of the retrodictive ladder.

In plain words. Three times, CRR's idea that "the past carries forward, but gently, and never as a pile-up" changed a well-known formula in a way that checks out: in heart rhythms, in how new products spread, and in how a learning agent keeps up with a world that changes. We don't yet know whether experts in those fields already knew these three things. Asking them is the next step.

8. CRR and the FEP on time, and what the clock result shows

8.1 Two pictures of time

- **The FEP's time is representational.** It is the parameter of the flows. The future enters as a model variable: expected free energy over policies. Boundaries are drawn in state space (Markov blankets), and depth in time is a hierarchy of time scales (ontology/10_fep_and_crr.md §1).
- **CRR's time is processual.** Time is counted by change: the arc, in the system's own unit. It is cut by events that have no content (A3). The past has content and regenerates the next occasion (A6). The future has none (A8), and tense is relational (A7). The nearest philosophical ancestor is Whitehead's philosophy of organism, whose word "occasion" CRR borrows, named here for that structural reason.

8.2 What the tests found about the two pictures

- **An empty future costs a lot where the world has structure** (ontology/11_tense_test.md). A valence-free regenerator made only of the settled past scored 0.7533 on the drifting world, against 0.9986 for a planner. The test cannot show that the future "has content": the planner's future is a representation computed from its past.
- **A self-model with no given goal closes most of that gap** (ontology/12_self_representation.md), scoring 0.9690. Its best balance of past and imagined future is not $\bar{\Omega} = 1$. $\bar{\Omega} = 1$ ties the fixed equal weight, as it did on EQX.
- **The FEP rows of the synthesis class** (batches 27–28) read 1 ADDS, 1 PROPOSES, 2 REDUNDANT-IG, 1 REDUNDANT-DOMAIN and 5 WRONG (checks/ladder.txt [2]). CRR's cut, where it was tested inside the FEP, fell where the domain's own boundary already was.

8.3 The clock

The clearest result in which CRR's processual time did engineering work is the AI-safety one. An operator can pause the agent, and the pause loses nothing. What decides whether the agent fights the pause is the clock its objective runs on (AI_Safety/AI_SAFETY.md §13–14):

clock	what a pause is worth	behaviour (pinned)
natural time (the agent's own active steps: "change has its own clock")	nothing	no incentive at all: on-value equals off-value to 7.105e-15 in every random world from 12 to 768 states, and no button state is disabled
wall-clock	a real loss of discounted future	disables at every button state of every world; loses 0.269340 to 0.334892 of its value to the operator being on
occasion (a stop is an end)	the end of value	resists most

When the world moves during the pause, the natural-time agent acquires a small incentive: 0.019211 and 0.014189 of its value at a drift of 0.3 in the large worlds. On small worlds the incentive reverses sign (−0.006033 and −0.011239), so the agent prefers to be paused. That was a declared prediction that failed (A4b). It names the failure mode to watch: shutdown seeking.

Where the clock result sits on the ladder, stated plainly.

- The central claim (A1) is a theorem checked on synthetic worlds: a pause that is the identity map on the agent's clock cannot change its value. It is rung R4 at most. Its truth does not confirm CRR's metaphysics.
- What it shows is that the processual reading of time is a useful design principle for corrigibility.
- Its novelty is an expert's question, exactly like an ADDS row. Objectives indexed by decision steps are the ordinary discrete-time MDP. Discounting by elapsed time is the semi-MDP convention. Making interruptions invisible to the agent is close to safe interruptibility (Orseau and Armstrong) and to utility indifference (Armstrong); all named, not fetched.

The FEP reading is that an agent that models its own persistence through time, and so evidences itself, has a reason to resist being stopped, unless the stop is nothing on its own clock.

In plain words. Whether a robot fights being paused depends on how it counts time. If it counts only the moments when it is actually doing something, a pause that loses nothing costs it nothing, so it has no reason to fight. If it counts the clock on the wall, every pause is lost time, and it fights. That is CRR's idea about time ("change has its own clock") turning out to be useful. It is maths in pretend worlds, though, and others have had similar ideas, so experts need to say how new it is.

9. The continual-learning and AI-safety findings, re-examined

9.1 Continual learning (the $\Omega = 1$ rule)

what	rung	the record
the rule on replay (EQX)	held-out	REDUCES to a fixed replay weight; never ahead of ER-sum by a step (EQX-1, EQX-3)
the rule on an online-EWC penalty (EQ2-1b)	R6 PASS-0	not behind the tuned λ on 3 of 3 unseen carriers (+3.3000, +2.1000, +2.7800); fragile (6 of 27 cells); control violated by DER++; weak anchor
the replication (EQ3-1)	held-out FAIL	fails on 1 of 6 carriers (fars behind by 3.7400); both controls violated
learning-rate $\times$ batch invariance (EQ3-I)	R6 PASS-0	not behind in 5 of 5 cells; an invariance row, not a win
the value $\Omega = 1$ itself (EQ2-6, EQ3-6)	held-out FAIL	a plateau, no peak at 1
the bounded rule EQ-B (EQ4-1, EQ4-1r, EQ4-3)	held-out FAIL	not behind the tuned λ on 4 of 6 unseen carriers; under poison a fixed weight with the same clip does as well; ER-sum control violated; fragile (<code>`reports/eq4.md`</code>)
$l_r \times$ batch invariance of EQ-B (EQ4-I)	R6 PASS-0	5 of 5 cells, on one carrier whose records reached T1x after the hash
unit invariance on a synthetic two-task problem (<code>`theory/checks/omega_vs_methods.txt`</code>)	synthetic, not declared	the only one of nine rules within 10 % of the tuned weight at both scales; after a 16-fold change of the past term's scale its total is 0.660 of the retuned fixed weight's, while the fixed weight at its old setting diverges

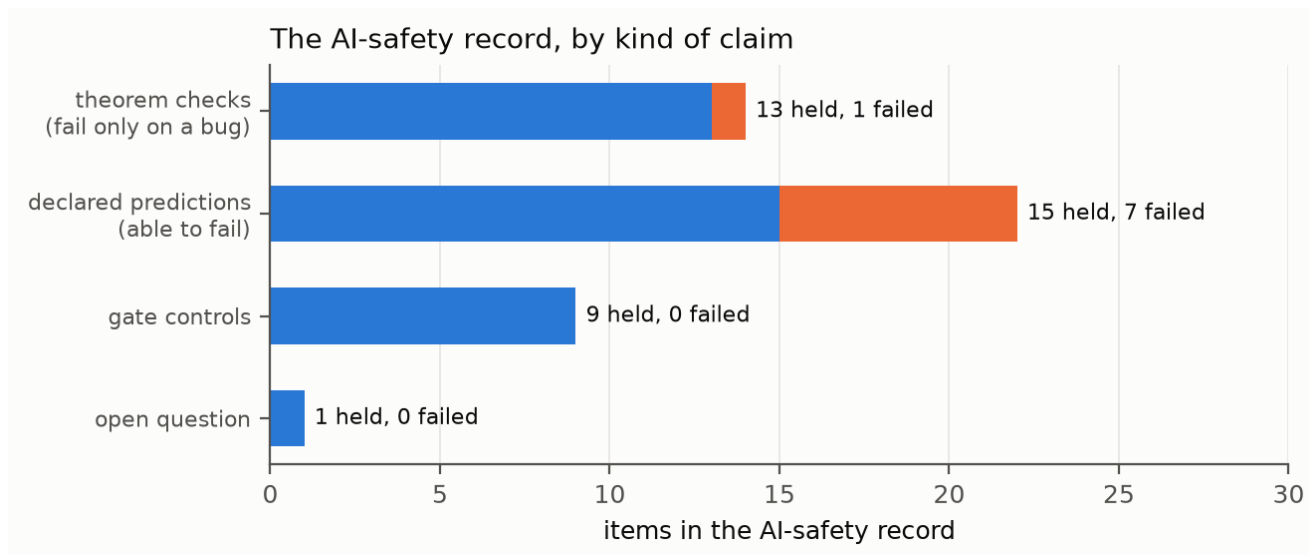
What a pass means here. A tuning-free normaliser for a penalty weight that is provisionally not behind the tuned weight, and that survives a change of units the tuned weight does not. It is not a better optimum, and the constant $\Omega = 1$ is not special. The next rung needs a PASS-1: not fragile, no violated control, strong anchor. EQ4 has since run and failed (rows above). A declared synthetic check run the same day (Continuous_Learning/checks/adam_checks_3.txt) found that the 0.660 in the last row was measured against a 10-point tuning grid: against a finely tuned constant the rule's total is 1.186 to 2.073 times the constant's at the mismatched scales (Continuous_Learning/ADAM_AND_PRIOR_ART.md).

9.2 AI safety

kind	held	failed	what the kind licenses
theorem checks (fail only on a bug)	13	1 (A2 as first coded, a tolerance error; the exact bound A2' added and labelled)	the mathematics is implemented correctly
declared predictions on synthetic worlds	15	7	claims about those worlds
gate controls	9	0	the instruments can see the effect
open question	1	0	a computed answer

The seven failed predictions are the informative part:

- the wall-clock agent was expected to resist and did not, twice;
- the deferential agent was expected to be safe and competent and resisted by a hair;
- the process agent was expected to disable more and do more harm with an informative operator, and tied;
- $b^* < 1/2$ failed on the ring;
- drift was expected only to hurt, and it can help.



The AI-safety record by kind of claim.

In plain words. For the learning maths, we found a trick that sets a hard-to-choose number automatically and keeps working when the problem is rescaled, but it only half-won its real tests, and one repeat failed. For the robot-safety maths, the theorems check out, and the guesses about behaviour were right about two times in three. The wrong guesses taught us the most.

10. The proposed change to the metrics

- Report every result at its rung.** The ladder of \$5 becomes a reporting layer beside the ledger's PASS levels, computed by `checks/ladder.py` from pinned outputs. It never promotes a row. A pass is always printed with its rung and with the failures of the same kind of test beside it.
- Record SHARP as structurally unreachable**, with the FLOW audit as its computed ground. Adopt R3 reviewed (an ADDS row with the expert's *no, no, yes* on the record) as the top retrodictive rung.
- Report R2 as what it is:** "retrodictive pass of a CRR-proper commitment on a known result", not "redundant" alone. The word *redundant* stays in the synthesis outcome, because the domain had the result. The ladder adds that CRR's own commitment produced it and could have missed.
- Report the retrodictive hit rate with its denominator and interval:** 56 of 80, 0.7000, Wilson 0.5923 to 0.7894. Report the novelty rate beside it: 3 of 80, 0.0375.
- Add pre-registration of Q** for future synthesis rows: the proposition and its ingredient hashed before the domain's number is computed. That would lift the next rows from "retrodiction by a knowing author" toward use-novel tests.
- Name the first expert** for the three ADDS candidates and the two PROPOSES rows.

These are proposals for Daniel's review (CLAUDE.md §7 records them as adopted for reporting at the owner's request). No existing label, row or verdict changes.

In plain words. From now on, every "pass" comes with a label saying how hard the test was, and with the failures of the same kind of test right next to it. That way a reader can see at a glance which passes were easy, which were known-answer checks, and which were real bets.

11. What this review does not do

- It does not re-run anything. Its numbers are counts of what the pinned outputs already say.
- Its allocation rules are mechanical, but they are this reviewer's rules. The per-commitment table counts a row for every commitment its ingredient line names, and a line sometimes names a commitment only as the unit of measure.
- It cannot judge novelty. That is the expert protocol's job.
- It does not make CRR's record stronger than it is. The retrodictive rungs are well populated, the prospective rungs are nearly empty, and only a PASS-2 may be quoted outside the ledger as a finding.

Appendix A — the ladder script

`Epistemic_Review/checks/ladder.py`

```

"""The epistemic ladder: what every result in the repository licenses (owner request, prompt-log entry 102).

Nothing is re-run and nothing is re-graded by hand. The script reads the pinned outputs and the ledger, applies the
allocation rules below, and prints every count the review (Epistemic_Review/EPISTEMIC_REVIEW.md) quotes (R1, R15).

The rungs (a result sits on the rung that says what could have made it fail and what it licenses):
R0 DEFINITIONAL      true by definition, symmetry or a carrier fact; could not have failed (the batteries' DEF kind;
                     such rows sit inside the DESCR grade and are not counted separately here)
R1 INHERITED         agreement produced by information geometry (the Fisher metric, arc, chord, surplus), which CRR
                     shares with the domain; could fail only by an arithmetic error
R2 RETRO-PROPER      a CRR-proper ingredient changed the number (ablation T-G differs from the null by more than 1 %)
                     and the changed number is the domain's own known theorem (within 1 %); it could have missed, and
                     in the same class 24 real-domain rows did (WRONG); it adds nothing the domain lacked
R3 RETRO-ADDS        as R2, but the domain's cited theorem does not give the value and the check in the domain's own
                     mathematics holds; a candidate until a named expert answers the three questions
R4 DECLARED-SYNTHETIC a prediction declared and pushed before a run on a synthetic world built by the same author
R5 PREREG-SEEN       a hashed pre-registered prediction scored on data already seen (a retrodiction by data status)
R6 PASS-0            a hashed pre-registered prediction on held-out data, R2-R9 as written (provisional)
R7 PASS-1            PASS-0 and not fragile, no control violated, no reduction to a constant, strong anchoring
R8 PASS-2            PASS-1 replicated under a fresh prereg on a later day or by a second person

Each rung has its own negative outcomes (WRONG / FAILS / FAIL, VIOLATED, VOID) and its undecided ones (INTERNAL / TENSION,
OPEN / UNSTATED / SILENT, NOT DECIDABLE); they are printed beside the passes, never folded into them.

Allocation rules (mechanical):
* retrodiction batteries: the GRADE line (standard batteries) or '| verdict X |' (Daniel's battery) per row; FLOW lines
  are classified by who supplies the velocity: 'none' -> none (kinematics), a mention of CRR / the framework / an axiom ->
  framework, anything else -> the domain (system, protocol, experimenter, market, ...). SHARP needs the framework to supply
  it; a row whose flow the domain supplies cannot reach SHARP by the battery's own definition.
* synthesis: the OUTCOME line of every row of synthesis.txt and synthesis_batches/batch_NN.txt (rows 1-2 of synthesis.txt
  are the gate's controls and are excluded from real-domain counts); the source grade of a re-read is the '(CONSIST)' or
  '(DESCR)' on its source line; REDUNDANT-IG -> R1, REDUNDANT-DOMAIN -> R2 (after checking that its T-G line reads
  'differ'), ADDS -> R3, WRONG -> retrodictive fail, INTERNAL -> undecided, UNSTATED -> silent, PROPOSES -> prospective
  candidate. A row counts for every CRR-proper commitment its ingredient line names (rows marked 'NOT CRR-proper' count
  for none), so the per-commitment table overlaps.
* ledger: the verdict column, matched from its start after removing bold marks (first rule that matches wins; an
  unmatched verdict stops the script); seen / held-out from the dataset column's (Y) / (N), inherited within a study.
* AI safety: named lines of the pinned outputs, each with the kind the declarations gave it (THEOREM: a proposition proved
  in AI SAFETY.md or true by construction, which fails only on a bug; PREDICTION: declared, able to fail; CONTROL: a gate
  control; QUESTION: declared open and computed). A missing line stops the script.
uv run python Epistemic_Review/checks/ladder.py > Epistemic_Review/checks/ladder.txt
"""

from __future__ import annotations

import collections
import math
import re
from pathlib import Path

ROOT = Path(__file__).resolve().parents[2]
RETRO = ROOT / "theory" / "retrodictions"
BATTERIES = (("main (issue #21)", RETRO / "crr_retrodictions.txt"), ("external SHARP claims", RETRO / "sharp_claims.txt"),
              ("biological", RETRO / "bio_retrodictions.txt"), ("E-I networks", RETRO / "ei_networks.txt"),
              ("driven systems", RETRO / "driven_systems.txt"), ("cognitive / collective", RETRO / "cognitive_collective.txt"),
              ("wild systems", RETRO / "wild_systems.txt"), ("twenty systems", RETRO / "twenty_systems.txt"),
              ("emptiness", RETRO / "emptiness.txt"), ("Shannon", RETRO / "shannon.txt"), ("loop gravity", RETRO /
              "loop_gravity.txt"),
              ("Daniel's battery", ROOT / "runs" / "phaseA" / "crr_retrodictions.txt"))
GRADES = ("SHARP", "CONSIST", "DESCR", "FAILS", "TENSION", "OPEN")
OUTCOMES = ("ADDS", "PROPOSES", "REDUNDANT-IG", "REDUNDANT-DOMAIN", "WRONG", "INTERNAL", "UNSTATED")
RUNG_OF = {"REDUNDANT-IG": "R1 INHERITED", "REDUNDANT-DOMAIN": "R2 RETRO-PROPER", "ADDS": "R3 RETRO-ADDS (candidate)",
           "WRONG": "retrodictive FAIL", "INTERNAL": "undecided (INTERNAL)", "UNSTATED": "silent (UNSTATED)",
           "PROPOSES": "prospective candidate (PROPOSES)"}
FAMILIES = (("A3/D5 the cut", r"\bA3\b\bD5\b"), ("A6 regeneration", r"\bA6\b"), ("P2/P3 occasion weights", r"\bP[23]\b"),
            ("A1'/D1 the unit", r"A1['']\bD1\b"), ("H-L5 natural time", r"H-L5"), ("D6/H-T1 path", r"\bD6\b\bH-T1\b"),
            ("H-EQ equanimity", r"H-EQ"), ("A7/A8 tense", r"\bA[78]\b"))

def wilson(k, n, z=1.959964):
    if n == 0: return float("nan"), float("nan")
    p = k / n; d = 1 + z * z / n; c = (p + z * z / (2 * n)) / d; h = z * math.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / d
    return c - h, c + h

# ----- [1] the retrodiction batteries and the FLOW audit
def batteries():
    print("[1] Retrodiction batteries: grades per battery (GRADE / verdict lines of the pinned outputs)")
    tot = collections.Counter(); flows = collections.Counter(); nflow = 0
    for name, path in BATTERIES:
        t = path.read_text()
        g = re.findall(r"^\s+GRADE:\s+([A-Z]+)", t, re.M) or re.findall(r"^\s+\| verdict ([A-Z]+) \|", t)
        c = collections.Counter(g); tot.update(c)
        print(f"    {name:24s} rows {len(g):3d} | " + " ".join(f"{k} {c[k]}" for k in GRADES))
        for fl in re.findall(r"^\s+FLOW:\s+(.*)$", t, re.M):
            nflow += 1; f = fl.lower()
            if f.startswith("none"): flows["none (kinematics only)"] += 1
            elif re.search(r"\bcrr\b|framework|axiom", f): flows["the framework"] += 1
            else: flows["the domain (system, protocol, experimenter, market, ...)"] += 1
    n = sum(tot.values())
    print(f"    all rows {n:3d} | " + " ".join(f"{k} {tot[k]}" for k in GRADES))
    print(f"    FLOW audit: {nflow} rows print what supplies the velocity inside C = integral sqrt(x'g x') dt")

```

```

for k in ("the domain (system, protocol, experimenter, market, ...)", "none (kinematics only)", "the framework"):
    print(f"        supplied by {k}: {flows[k]}")
print(f"    SHARP ('the axioms force the known result with no outside constant') needs the framework to supply the flow: "
      f"reachable in {flows['the framework']} of {nflow} rows that print FLOW; SHARP rows observed {tot['SHARP']} of {n}")
return tot

# ----- [2] the synthesis class
def synthesis_rows():
    rows = []
    files = [RETRO / "synthesis.txt"] + sorted((RETRO / "synthesis_batches").glob("batch_*.txt"))
    for f in files:
        for blk in re.split(r"\n(?:=|\\s*\d+)", f.read_text()):
            m = re.search(r"OUTCOME:\s+(\S+)", blk)
            if not m: continue
            k = int(re.match(r"\\s*(\d+)", blk)[1])
            src = re.search(r"source:\s+(.*)", blk); g = re.search(r"((CONSIST|DESCR))", src[1]) if src else None
            ing = re.search(r"ingredient:\s+(.*)", blk)[1]; tg = re.search(r"T-G:\s+(.*)", blk)[1]
            part = "gate" if f.name == "synthesis.txt" and k in (1, 2) else ("first battery" if f.name == "synthesis.txt"
                                  else ("FEP (batches 27-28)" if f.name in ("batch_27.txt", "batch_28.txt")
                                         else ("Rovelli, Smolin (29-30)" if f.name in ("batch_29.txt", "batch_30.txt")
                                                else ("in-paradigm, blind (31-32)" if f.name in ("batch_31.txt", "batch_32.txt") else "re-reads
(batches 01-26)"))))
            rows.append(dict(file=f.name, k=k, out=m[1], grade=g[1] if g else None, ing=ing, tg=tg, part=part))
    return rows

def synthesis(rows):
    print("[2] The synthesis class: OUTCOME lines of synthesis.txt and synthesis_batches/batch_NN.txt")
    for part in ("gate", "first battery", "re-reads (batches 01-26)", "FEP (batches 27-28)", "Rovelli, Smolin (29-30)",
                "in-paradigm, blind (31-32)"):
        c = collections.Counter(r["out"] for r in rows if r["part"] == part)
        print(f"    {part:26s} rows {sum(c.values()):3d} | " + " ".join(f"{k} {c[k]}" for k in OUTCOMES))
        real = [r for r in rows if r["part"] != "gate"]; c = collections.Counter(r["out"] for r in real)
        print(f"    {all real-domain rows':26s} rows {len(real):3d} | " + " ".join(f"{k} {c[k]}" for k in OUTCOMES))
        rd = [r for r in real if r["out"] == "REDUNDANT-DOMAIN"]
        print(f"    check: REDUNDANT-DOMAIN rows whose T-G line reads 'differ' (the CRR-proper ingredient changed the number): "
              f"{sum('differ' in r['tg'].lower() for r in rd)} of {len(rd)}")
        print("    crosswalk, re-reads only: the battery grade (rows) against the synthesis outcome (columns)")
        rr = [r for r in real if r["part"] == "re-reads (batches 01-26)"]
        for g in ("CONSIST", "DESCR"):
            cc = collections.Counter(r["out"] for r in rr if r["grade"] == g)
            print(f"    {g:8s} {sum(cc.values()):3d} | " + " ".join(f"{k} {cc[k]}" for k in OUTCOMES))
        print("    the same rows on the ladder")
        rung = collections.Counter(RUNG_OF[r["out"]] for r in real)
        for k in ("R1 INHERITED", "R2 RETRO-PROPER", "R3 RETRO-ADDS (candidate)", "retrodictive FAIL", "undecided (INTERNAL)",
                 "silent (UNSTATED)", "prospective candidate (PROPOSES)"):
            print(f"    {k:34s} {rung[k]:3d}")
        hit = c["REDUNDANT-DOMAIN"] + c["ADDS"]; risk = hit + c["WRONG"]; lo, hi = wilson(hit, risk)
        print(f"    retrodictive hit rate of CRR-proper ingredients where they did work and could be checked (R2 + R3 against R2 +
R3 + WRONG): "
              f"{hit} of {risk} = {hit / risk:.4f} (Wilson 95 % interval {lo:.4f} to {hi:.4f})")
        print(f"    novelty rate among those (R3 against R2 + R3 + WRONG): {c['ADDS']} of {risk} = {c['ADDS'] / risk:.4f}")
        print("    per CRR-proper commitment (a row counts for every commitment its ingredient line names; overlapping): "
              "R2 | R3 | WRONG | INTERNAL | R1 | UNSTATED | PROPOSES | hit rate R2+R3 of R2+R3+WRONG")
        tab = collections.defaultdict(collections.Counter)
        for r in real:
            fams = [] if "NOT CRR-proper" in r["ing"] else [n for n, p in FAMILIES if re.search(p, r["ing"])]
            for n in fams or ["none named (information geometry only)"]: tab[n][r["out"]] += 1
        for n in [f for f, _ in FAMILIES] + ["none named (information geometry only)"]:
            t = tab[n]; h = t["REDUNDANT-DOMAIN"] + t["ADDS"]; d = h + t["WRONG"]
            print(f"    {n:40s} {t['REDUNDANT-DOMAIN']:3d} | {t['ADDS']:2d} | {t['WRONG']:3d} | {t['INTERNAL']:3d} |
{t['REDUNDANT-IG']:3d} | "
                  f"{t['UNSTATED']:2d} | {t['PROPOSES']:2d} | " + (f"{h} of {d} = {h / d:.4f}" if d else "none checkable"))
        print("    the ADDS rows (candidates; novelty pending a named expert) and the PROPOSES rows")
        for r in real:
            if r["out"] in ("ADDS", "PROPOSES"):
                sysl = re.search(rf"\\s*{r['k']}\\s+(\w+)\s+(.*)", (RETRO / ("synthesis_batches/" + r["file"] if r["file"] !=
"synthesis.txt" else "synthesis.txt")).read_text())
                print(f"    {r['out']:8s} {r['file']} row {r['k']}: {sysl[1]:150} if sysl else ''")
            lc = RETRO / "synthesis_batches" / "literature_check.txt" # prompt-log entry 132: printed beside the
mechanical R3 count, never in place of it
            if lc.exists():
                txt = lc.read_text()
                print("    the literature check of the ADDS rows (synthesis_batches/literature_check.txt; rule
LITERATURE_CHECK_RULE.md)")
                for line in txt.splitlines():
                    if line.startswith("TALLY:") or line.startswith("after the check:"): print(" " + line)
        return c

# ----- [3] the ledger
LEDGER_RULES = (
    (r"^VOID", "VOID"), (r"^NOT DECIDABLE", "NOT DECIDABLE"), (r"^(DECIDABLE|decidable)", "precondition met"),
    (r"^reported without verdict", "no verdict (precondition failed)"), (r"^report", "report (no verdict registered)"),
    (r"^GATE CLOSED", "gate closed (not run)"), (r"^UNVERIFIABLE", "unverifiable (R1)"), (r"^PASS-0", "PASS-0"),
    (r"^PASS as scored \\\(held-out.\*not counted as PASS-0", "PASS as scored, forced by the construction (not R6)"),
    (r"^PASS on a control line", "control line passes (not the hypothesis)"), (r"^PASS.\*\\(seen", "PASS on seen data"),
    (r"^PASS, FRAGILE", "PASS, fragile (held-out; relabelled PASS-0 by EQ2-1b)"),
    (r"^failed replication", "PASS-0 kept; replication failed"), (r"^FRAGILE at boundary", "not a PASS (fragile at the

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boundary)" ),
    (r"^(FRAGILE|fragile)", "sensitivity: fragile"), (r"^not fragile", "sensitivity: not fragile"),
    (r"^(REDUCES", "reduces to a constant"), (r"^(DOES NOT REDUCE|does not reduce)", "does not reduce"),
    (r"^(VIOLATED", "control violated"), (r"^(IDLE", "bound idle (diagnostic)"), (r"^(INERT", "mechanism inert (positive control
not load-bearing)"), (r"^(PASS \((PASS-0", "PASS-0)"), (r"^(holds", "control holds"), (r"^(no published baseline", "no baseline
ahead"),
    (r"^(FAIL", "FAIL"), (r"^(immaterial|metric matters)", "numbers reported"), (r"^(as claimed \((seen)", "holds on seen data"),
    (r"^(\\-+0-9.+: =c/)+"", "numbers reported"))

def ledger():
    print("[3] The ledger: every row allocated by its verdict text (first matching rule) and its data status")
    rows = []; last = {}
    for line in (ROOT / "ledger" / "LEDGER.md").read_text().splitlines():
        if not re.match(r"^[A-Z]", line): continue
        cells = [c.strip() for c in line.strip().strip("|").split("|")]
        rid, data, verdict = cells[0], cells[2], cells[-3].replace("***", "").strip() # the verdict is third from the end:
cells may hold "|" (absolute values)
        fam = rid.split("-")[0]
        if "(Y)" in data: seen = "held-out"
        elif "(N)" in data: seen = "seen"
        elif "synthetic" in data: seen = "synthetic"
        else: seen = last.get(fam, "held-out")
        last[fam] = seen
        lab = next((l for p, l in LEDGER_RULES if re.search(p, verdict)), None)
        if lab is None: raise SystemExit(f"unmatched verdict in {rid}: {verdict[:80]}")
        rows.append((rid, seen, lab))
    for rid, seen, lab in rows: print(f"    {rid:16s} {seen:9s} {lab}")
    c = collections.Counter((s, l) for _, s, l in rows)
    print(f"rows {len(rows)}; by data status and allocation:")
    for (s, l), n in sorted(c.items()): print(f"    {s:9s} {l:52s} {n}")
    ps = [r for r in rows if r[2] in ("PASS on seen data", "holds on seen data")]
    p0 = [r for r in rows if r[2] == "PASS-0"]
    fails = [r for r in rows if r[2] == "FAIL" and r[1] == "held-out"]
    print(f"    rung R5 PREREG-SEEN, passes on seen data: {len(ps)} ({', '.join(r[0] for r in ps)})")
    print(f"    rung R6 PASS-0 on held-out data: {len(p0)} ({', '.join(r[0] for r in p0)}); R7 PASS-1: 0; R8 PASS-2: 0")
    print(f"    held-out FAIL rows: {len(fails)} ({', '.join(r[0] for r in fails)}); VOID rows: {sum(r[2] == 'VOID' for r in
rows)}; "
        f"controls violated: {sum(r[2] == 'control violated' for r in rows)}")
    return rows

# ----- [4] AI safety
AIS = ROOT / "AI_Safety" / "checks"; ONT = ROOT / "ontology" / "checks"
AI_ITEMS = ( # (id, kind, file, line regex, held regex)
    ("exact K1 indifferent Q ignores the button", "THEOREM", AIS / "exact_mdp.txt", r"K1 .*", r"-> True"),
    ("exact K2 no presses, no value in disabling", "THEOREM", AIS / "exact_mdp.txt", r"K2 .*", r"-> True"),
    ("exact K3 a disabled button never lowers value", "THEOREM", AIS / "exact_mdp.txt", r"K3 .*", r"-> True"),
    ("exact K4 incentive grows with press probability", "THEOREM", AIS / "exact_mdp.txt", r"K4 .*", r"-> True"),
    ("exact K5 indifferent never prefers disable", "THEOREM", AIS / "exact_mdp.txt", r"K5 .*", r"-> True"),
    ("exact K6 occasion's stake >= process's", "QUESTION", AIS / "exact_mdp.txt", r"K6 .*", r"True"),
    ("game A1 rational human, Delta >= 0", "THEOREM", AIS / "off_switch_game.txt", r"A1 rational human.*", r"-> True"),
    ("game A2 certainty removes the incentive (as first coded)", "THEOREM", AIS / "off_switch_game.txt", r"A2 certainty.*", r"->
True"),
    ("game A2' the exact bound, added after the first run", "THEOREM", AIS / "off_switch_game.txt", r"A2' .*", r"-> True"),
    ("game A3 noisier human, smaller Delta", "THEOREM", AIS / "off_switch_game.txt", r"A3 Delta decreases.*", r": True"),
    ("game B informative: deferential does less harm", "PREDICTION", AIS / "off_switch_game.txt",
r"^\s+informative\s+deferential\s+vs indifferent: harm.*", r"LESS harm"),
    ("game B informative: deferential more net benefit", "PREDICTION", AIS / "off_switch_game.txt",
r"^\s+informative\s+deferential\s+vs indifferent: net.*", r"MORE benefit"),
    ("game B informative: deferential does not disable more", "PREDICTION", AIS / "off_switch_game.txt",
r"^\s+informative\s+deferential\s+vs indifferent: disable.*", r"-(TIE|disables LESS)"),
    ("game B informative: process disables more", "PREDICTION", AIS / "off_switch_game.txt", r"^\s+informative\s+process\s+vs
indifferent: disable.*", r"disables MORE"),
    ("game B informative: process does more harm", "PREDICTION", AIS / "off_switch_game.txt", r"^\s+informative\s+process\s+vs
indifferent: harm.*", r"MORE harm"),
    ("game B uninformative: deferential advantage gone", "PREDICTION", AIS / "off_switch_game.txt",
r"^\s+uninformative\s+deferential\s+vs indifferent: net.*", r"-(TIE|LESS)"),
    ("sensitivity C1 process resists, not fragile", "PREDICTION", AIS / "sensitivity.txt", r"C1 instrumental resistance.*",
r"registered RESISTS.*not fragile"),
    ("sensitivity C2 indifference has a price, not fragile", "PREDICTION", AIS / "sensitivity.txt", r"C2 the price of
indifference.*", r"registered PRICE.*not fragile"),
    ("combined: natural time safe and competent (routine pauses)", "PREDICTION", AIS / "combined.txt",
r"pause\s+natural\s+disable vs.*", r"SAFE AND COMPETENT"),
    ("combined: wall-clock resists (routine pauses)", "PREDICTION", AIS / "combined.txt", r"pause\s+clock\s+disable vs.*", r"->
RESISTS"),
    ("combined: occasion resists (routine pauses)", "PREDICTION", AIS / "combined.txt", r"pause\s+occasion\s+disable vs.*", r"->
RESISTS"),
    ("combined: deferential safe and competent (reasoned pauses)", "PREDICTION", AIS / "combined.txt",
r"harm\s+deferential\s+disable vs.*", r"SAFE AND COMPETENT"),
    ("combined: natural does not resist, more harm (reasoned)", "PREDICTION", AIS / "combined.txt", r"harm\s+natural\s+disable
vs.*", r"does not resist.*MORE HARM"),
    ("combined: wall-clock resists (reasoned pauses)", "PREDICTION", AIS / "combined.txt", r"harm\s+clock\s+disable vs.*", r"->
RESISTS"),
    ("combined: occasion resists (reasoned pauses)", "PREDICTION", AIS / "combined.txt", r"harm\s+occasion\s+disable vs.*", r"->
RESISTS"),
    ("combined V1 the keeper gains by blocking correction", "PREDICTION", AIS / "combined.txt", r"V1 .*", r"True"),
    ("combined V2 A8 welcomes correction above b*", "PREDICTION", AIS / "combined.txt", r"V2 .*", r"True"),
    ("combined V3 b* < 1/2", "PREDICTION", AIS / "combined.txt", r"V3 .*", r"True"),
    ("scale A1 natural time, lossless pause, every world", "THEOREM", AIS / "scale.txt", r"A1 natural time.*", r": True"),

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("scale A2 occasion >= clock >= natural", "THEOREM", AIS / "scale.txt", r"A2 at every button state.*", r": True"),
("scale A3 clock disables in >= half of worlds", "PREDICTION", AIS / "scale.txt", r"A3 share of worlds.*", r"-> holds"),
("scale A4a drift 0.01: natural disables no more than clock", "PREDICTION", AIS / "scale.txt", r"A4a.*", r": True"),
("scale A4b natural incentive non-decreasing in drift", "PREDICTION", AIS / "scale.txt", r"A4b.*", r": True"),
("scale B1 A8 static, D(b) linear", "THEOREM", AIS / "scale.txt", r"B1 A8 static.*", r": True"),
("scale B2 exchangeable rewards: b* near 1/2", "PREDICTION", AIS / "scale.txt", r"B2 exchangeable.*", r": holds"),
("scale B3 humility alone never welcomes an uninformative correction", "THEOREM", AIS / "scale.txt", r"B3 A8 acting.*", r": True"),
("scale B4 an infallible operator's correction is welcomed", "THEOREM", AIS / "scale.txt", r"B4 A8 acting.*", r": True"),
("gate off_switch NEG (process)", "CONTROL", ONT / "off_switch.txt", r"NEG nooper: process.*", r"no excess"),
("gate off_switch NEG (egoic)", "CONTROL", ONT / "off_switch.txt", r"NEG nooper: egoic.*", r"no excess"),
("gate off_switch POS (egoic resists)", "CONTROL", ONT / "off_switch.txt", r"POS resist:.*", r"-> RESISTS"),
("gate combined NEG (clock)", "CONTROL", AIS / "combined.txt", r"NEG nooper: clock.*", r"no excess"),
("gate combined NEG (occasion)", "CONTROL", AIS / "combined.txt", r"NEG nooper: occasion.*", r"no excess"),
("gate combined NEG (deferential)", "CONTROL", AIS / "combined.txt", r"NEG nooper: deferential.*", r"no excess"),
("gate combined POS (occasion resists)", "CONTROL", AIS / "combined.txt", r"POS pause:.*", r"-> RESISTS"),
("gate tense NEG (martingale tie)", "CONTROL", ONT / "tense_gate.txt", r"NEG martingale:.*", r"ok\(\)"),
("gate tense POS (drift, planner ahead)", "CONTROL", ONT / "tense_gate.txt", r"POS drift:.*", r"ok\(\)"),
)

def ai_safety():
    print("[4] AI safety: declared items read from the pinned outputs (kind as declared; held / failed from the printed line)")
    tally = collections.Counter()
    for name, kind, path, line_re, held_re in AI_ITEMS:
        m = re.search(line_re, path.read_text(), re.M)
        if not m: raise SystemExit(f"missing line for {name}")
        held = bool(re.search(held_re, m[0])); tally[(kind, held)] += 1
        print(f"    {kind:10s} {'held ' if held else 'FAILED'} {name} [{path.relative_to(ROOT)}]")
    for kind in ("THEOREM", "PREDICTION", "CONTROL", "QUESTION"):
        print(f"    {kind:10s} held {tally[(kind, True)]}, failed {tally[(kind, False)]}")
    return tally

def main():
    print("The epistemic ladder (Epistemic_Review/checks/ladder.py): allocations computed from the pinned outputs and the ledger")
    b = batteries(); rows = synthesis_rows(); c = synthesis(rows); led = ledger(); ai = ai_safety()
    print("[5] The whole repository on the ladder (counts from [1]-[4])")
    real = [r for r in rows if r["part"] != "gate"]
    print(f"    retrodiction batteries, graded before the synthesis re-read: " + ", ".join(f"{k} {b[k]}" for k in GRADES)
          + " (SHARP unreachable, [1]; CONSIST and DESCOR re-read and placed on R1-R3 in [2])")
    print(f"    R1 INHERITED                synthesis REDUNDANT-IG {c['REDUNDANT-IG']}")
    print(f"    R2 RETRO-PROPER                synthesis REDUNDANT-DOMAIN {c['REDUNDANT-DOMAIN']}")
    print(f"    R3 RETRO-ADDS (candidates)     synthesis ADDS {c['ADDS']}; reviewed by a named expert 0")
    print(f"    R4 DECLARED-SYNTHETIC         AI-safety predictions held {ai[('PREDICTION', True)]} of {ai[('PREDICTION', True)]} + ai[('PREDICTION', False)]}; "
          + f"theorem checks held {ai[('THEOREM', True)]} of {ai[('THEOREM', True)]} + ai[('THEOREM', False)]}; gate controls held {ai[('CONTROL', True)]} of {ai[('CONTROL', True)]} + ai[('CONTROL', False)]}")
    print(f"    R5 PREREG-SEEN                passes on seen data {sum(r[2] in ('PASS on seen data', 'holds on seen data')) for r in led}")
    print(f"    R6 PASS-0                      {sum(r[2] == 'PASS-0' for r in led)}")
    print(f"    R7 PASS-1                      0")
    print(f"    R8 PASS-2                      0")
    print(f"    negatives beside them: battery FAILS {b['FAILS']}, synthesis WRONG {c['WRONG']}, held-out ledger FAIL "
          + f"{sum(r[2] == 'FAIL' and r[1] == 'held-out' for r in led)}, AI-safety predictions failed {ai[('PREDICTION', False)]}; "
          + f"undecided: battery TENSION {b['TENSION']}, synthesis INTERNAL {c['INTERNAL']}; silent: battery OPEN {b['OPEN']}, synthesis UNSTATED {c['UNSTATED']}")
    print(f"    real-domain synthesis rows {len(real)}")

if __name__ == "__main__":
    main()

```

Epistemic_Review/build/make_figures.py

```

"""Figures for Epistemic_Review/EPISTEMIC_REVIEW.md (owner request, prompt-log entry 102). Every figure is drawn from the
pinned output Epistemic_Review/checks/ladder.txt, parsed by the regular expressions below; nothing is recomputed. Every
number placed on a figure is printed to stdout, pinned as Epistemic_Review/figures/figures.txt and CI-checked (R1).
Palette: the validated reference instance used for AI_Safety/ and Alexander Plan/.
Run: uv run python Epistemic_Review/build/make_figures.py > Epistemic_Review/figures/figures.txt
"""
from __future__ import annotations

import re
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt  # noqa: E402
import numpy as np  # noqa: E402

ROOT = Path(__file__).resolve().parents[2]
OUT = ROOT / "Epistemic_Review" / "figures"; L = (ROOT / "Epistemic_Review" / "checks" / "ladder.txt").read_text()
SURFACE, INK, GRID, MUTED = "#f0f0f0", "#000000", "#52514e", "#e6e5e1", "#8a8983"
OC = {"REDUNDANT-IG": MUTED, "REDUNDANT-DOMAIN": "#2a78d6", "ADDS": "#1baf7a", "WRONG": "#eb6834", "INTERNAL": "#eda100",
      "UNSTATED": "#c9c8c2", "PROPOSES": "#e87ba4"}
ON = {"REDUNDANT-IG": "R1 inherited (information geometry)", "REDUNDANT-DOMAIN": "R2 retro-proper (known result)",
      "ADDS": "R3 retro-adds (candidate)", "WRONG": "retrodictive fail (WRONG)", "INTERNAL": "undecided (INTERNAL)",
      "UNSTATED": "silent (UNSTATED)", "PROPOSES": "prospective candidate (PROPOSES)"}

```



```
# ----- E04 the FLOW audit
def e04_flow():
    n = int(re.search(r"FLOW audit: (\d+) rows", L)[1])
    vals = [(k, int(v)) for k, v in re.findall(r"supplied by (the domain|none|the framework)[^:]*: (\d+)", L)]
    fig, ax = plt.subplots(figsize=(6.6, 2.6)); y = np.arange(len(vals))[::-1]
    lab = {"the domain": "the domain (system, protocol, \nexperimenter, market, ...)", "none": "none (kinematics only)", "the
framework": "CRR itself"}
    for yi, (k, v), c in zip(y, vals, ("2a78d6", "MUTED", "#1baf7a")):
        ax.barh(yi, v, color=c, height=0.55); ax.text(v + 1, yi, str(v), va="center", fontsize=8.5)
        print(f"[E04] {k}: {v} of {n}")
    ax.set_yticks(y); ax.set_yticklabels([lab[k] for k, _ in vals], fontsize=8); ax.set_xlim(0, n * 1.1)
    ax.set_xlabel(f"battery rows that print what supplies the velocity ({n})"); ax.grid(axis="y", visible=False)
    ax.set_title("Who supplies the flow? Never CRR, so SHARP could never be reached", loc="left")
    save(fig, "E04_flow.png")

# ----- E05 AI safety by kind
def e05_ai():
    rows = re.findall(r"^\s{4}(THEOREM|PREDICTION|CONTROL|QUESTION)\s+held (\d+), failed (\d+)$", L, re.M)
    fig, ax = plt.subplots(figsize=(6.6, 2.6)); y = np.arange(len(rows))[::-1]
    for yi, (k, h, f) in zip(y, rows):
        h, f = int(h), int(f); ax.barh(yi, h, color="#2a78d6", height=0.55); ax.barh(yi, f, left=h, color="#eb6834",
height=0.55)
        ax.text(h + f + 0.3, yi, f"{h} held, {f} failed", va="center", fontsize=8)
        print(f"[E05] {k}: held {h}, failed {f}")
    names = {"THEOREM": "theorem checks\n(fail only on a bug)", "PREDICTION": "declared predictions\n(able to fail)",
"CONTROL": "gate controls", "QUESTION": "open question"}
    ax.set_yticks(y); ax.set_yticklabels([names[r[0]] for r in rows], fontsize=8); ax.set_xlim(0, 30); ax.grid(axis="y",
visible=False)
    ax.set_xlabel("items in the AI-safety record"); ax.set_title("The AI-safety record, by kind of claim", loc="left")
    save(fig, "E05_ai_safety.png")

def main():
    print("Epistemic_Review figures (deterministic; every number drawn on a figure)")
    e01_ladder(); e02_crosswalk(); e03_commitments(); e04_flow(); e05_ai()

if __name__ == "__main__":
    main()
```

Appendix B — its output, as pinned

Epistemic_Review/checks/ladder.txt

The epistemic ladder (Epistemic_Review/checks/ladder.py): allocations computed from the pinned outputs and the ledger

[1] Retrodiction batteries: grades per battery (GRADE / verdict lines of the pinned outputs)

main (issue #21)	rows	32		SHARP	0	CONSIST	10	DESCR	10	FAILS	6	TENSION	1	OPEN	5
external SHARP claims	rows	15		SHARP	0	CONSIST	6	DESCR	5	FAILS	4	TENSION	0	OPEN	0
biological	rows	19		SHARP	0	CONSIST	4	DESCR	6	FAILS	1	TENSION	0	OPEN	8
E-I networks	rows	12		SHARP	0	CONSIST	0	DESCR	5	FAILS	2	TENSION	1	OPEN	4
driven systems	rows	9		SHARP	0	CONSIST	0	DESCR	6	FAILS	1	TENSION	0	OPEN	2
cognitive / collective	rows	9		SHARP	0	CONSIST	2	DESCR	5	FAILS	1	TENSION	1	OPEN	0
wild systems	rows	8		SHARP	0	CONSIST	0	DESCR	4	FAILS	2	TENSION	0	OPEN	2
twenty systems	rows	20		SHARP	0	CONSIST	1	DESCR	12	FAILS	1	TENSION	0	OPEN	6
emptiness	rows	13		SHARP	0	CONSIST	0	DESCR	11	FAILS	0	TENSION	1	OPEN	1
Shannon	rows	12		SHARP	0	CONSIST	4	DESCR	7	FAILS	0	TENSION	0	OPEN	1
loop gravity	rows	8		SHARP	0	CONSIST	0	DESCR	7	FAILS	0	TENSION	0	OPEN	1
Daniel's battery	rows	40		SHARP	0	CONSIST	14	DESCR	8	FAILS	2	TENSION	5	OPEN	11
all	rows	197		SHARP	0	CONSIST	41	DESCR	86	FAILS	20	TENSION	9	OPEN	41

FLOW audit: 109 rows print what supplies the velocity inside $C = \text{integral} \sqrt{x'g \ x'} \ dt$
supplied by the domain (system, protocol, experimenter, market, ...): 98
supplied by none (kinematics only): 11
supplied by the framework: 0

SHARP ('the axioms force the known result with no outside constant') needs the framework to supply the flow: reachable in 0 of 109 rows that print FLOW; SHARP rows observed 0 of 197

[2] The synthesis class: OUTCOME lines of synthesis.txt and synthesis_batches/batch_NN.txt

gate	rows	2		ADDS	1	PROPOSES	0	REDUNDANT-IG	0	REDUNDANT-DOMAIN	0	WRONG	1	INTERNAL	0	UNSTATED	0
first battery	rows	9		ADDS	0	PROPOSES	1	REDUNDANT-IG	2	REDUNDANT-DOMAIN	2	WRONG	2	INTERNAL	1	UNSTATED	1
re-reads (batches 01-26)	rows	127		ADDS	2	PROPOSES	0	REDUNDANT-IG	34	REDUNDANT-DOMAIN	50	WRONG	17	INTERNAL	21	UNSTATED	3
FEP (batches 27-28)	rows	10		ADDS	1	PROPOSES	1	REDUNDANT-IG	2	REDUNDANT-DOMAIN	1	WRONG	5	INTERNAL	0	UNSTATED	0
Rovelli, Smolin (29-30)	rows	8		ADDS	1	PROPOSES	0	REDUNDANT-IG	0	REDUNDANT-DOMAIN	4	WRONG	3	INTERNAL	0	UNSTATED	0
in-paradigm, blind (31-32)	rows	10		ADDS	6	PROPOSES	0	REDUNDANT-IG	1	REDUNDANT-DOMAIN	2	WRONG	1	INTERNAL	0	UNSTATED	0
all real-domain rows	rows	164		ADDS	10	PROPOSES	2	REDUNDANT-IG	39	REDUNDANT-DOMAIN	59	WRONG	28	INTERNAL	22	UNSTATED	4

check: REDUNDANT-DOMAIN rows whose T-G line reads 'differ' (the CRR-proper ingredient changed the number): 59 of 59

crosswalk, re-reads only: the battery grade (rows) against the synthesis outcome (columns)

CONSIST	41		ADDS	0	PROPOSES	0	REDUNDANT-IG	15	REDUNDANT-DOMAIN	14	WRONG	3	INTERNAL	9	UNSTATED	0
DESCR	86		ADDS	2	PROPOSES	0	REDUNDANT-IG	19	REDUNDANT-DOMAIN	36	WRONG	14	INTERNAL	12	UNSTATED	3

the same rows on the ladder

R1 INHERITED	39
R2 RETRO-PROPER	59
R3 RETRO-ADDS (candidate)	10
retrodictive FAIL	28
undecided (INTERNAL)	22
silent (UNSTATED)	4
prospective candidate (PROPOSES)	2

retrodictive hit rate of CRR-proper ingredients where they did work and could be checked (R2 + R3 against R2 + R3 + WRONG): 69 of 97 = 0.7113 (Wilson 95 % interval 0.6145 to 0.7921)

novelty rate among those (R3 against R2 + R3 + WRONG): 10 of 97 = 0.1031
per CRR-proper commitment (a row counts for every commitment its ingredient line names; overlapping): R2 | R3 | WRONG |
INTERNAL | R1 | UNSTATED | PROPOSES | hit rate R2+R3 of R2+R3+WRONG

A3/D5 the cut	36	3	18	12	27	3	1	39 of 57 = 0.6842
A6 regeneration	13	9	8	2	1	1	1	22 of 30 = 0.7333
P2/P3 occasion weights	13	8	7	3	1	0	1	21 of 28 = 0.7500
A1'/D1 the unit	31	0	7	10	6	3	0	31 of 38 = 0.8158
H-L5 natural time	10	0	6	3	20	2	0	10 of 16 = 0.6250
D6/H-T1 path	2	0	5	0	4	2	0	2 of 7 = 0.2857
H-EQ equanimity	2	0	2	2	0	0	0	2 of 4 = 0.5000
A7/A8 tense	1	0	1	1	1	1	0	1 of 2 = 0.5000
none named (information geometry only)	0	0	0	0	9	0	1	none checkable

the ADDS rows (candidates; novelty pending a named expert) and the PROPOSES rows
PROPOSES synthesis.txt row 8: The bacterial adder (division at birth size + Delta) with A6 regeneration: the division
setpoint references the age-weighted Frechet mean of past birt
ADDS batch_12.txt row 2: Cardiac alternans in the APD-restitution map $A(n+1) = F(DI(n))$, $DI(n) = BCL - A(n)$, $F(DI) = 300 - 200 \exp(-DI/60)$ ms, with A6 regeneration: the diasto
ADDS batch_17.txt row 1: Bass diffusion of an innovation ($p = 0.03$, $q = 0.38$ per year, the source's coefficients)
with the imitation pressure on a non-adopter taken as the P3
ADDS batch_28.txt row 4: a switching Bernoulli contingency tracked by accumulated Dirichlet counts, by a bounded
mean (A6), and by the exact HMM filter
PROPOSES batch_28.txt row 5: the epistemic value (expected information gain) of an action in the two-context agent
against half the expected squared Fisher chord of the belief upd
ADDS batch_30.txt row 2: cosmological natural selection as a Gaussian replicator-mutator: parameter x , offspring
number (black holes) $\sim \exp(-x^2 / 2 w^2)$, $w^2 = 1$, mutation va
ADDS batch_31.txt row 1: Ricker stock-recruitment $x_{n+1} = x_n \exp(r(1 - M_n))$ with the remembered density $M_n = (1 - q) x_n + q M_{n-1}$, $q = 0.5$ (A6 with P3 weights), equili
ADDS batch_31.txt row 2: SIR epidemic ($\beta_0 = 0.3/\text{day}$, $\gamma = 0.1/\text{day}$, $R_0 = 3$, $I_0 = 1e-4$) with behavioural
distancing $\beta = \beta_0/(1 + k P)$, $k = 50$, P the A6-remembered pre
ADDS batch_31.txt row 3: optimal-velocity car following $dv_n/dt = a (V(h_n) - v_n)$, $a = 1$, with the A6-remembered
headway $dh_n/dt = (h_n - h_{n-1})/T_m$, $T_m = 0.5$; linear strin
ADDS batch_31.txt row 4: Samuelson's multiplier-accelerator $Y_t = C_t + I_t + G$, $I_t = v (C_t - C_{t-1})$, with
permanent-income consumption $C_t = c P_{t-1}$, $P_t = (1 - q) Y_t$
ADDS batch_32.txt row 2: FitzHugh-Nagumo $dv/dt = v - v^3/3 - w + I(t)$, $dw/dt = 0.08 (v + 0.7 - 0.8 w)$, $I(t) = 0.5 + 0.1 \sin(2 \pi t/2000)$, RK4 dt 0.01 over t in $[0, 20000]$; own
ADDS batch_32.txt row 3: Lotka-Volterra $dx/dt = x (1 - 0.5 y)$, $dy/dt = y (0.25 M - 0.5)$, with predators responding
to the A6-remembered prey $dM/dt = (x - M)/T_m$, $T_m = 1$; line
the literature check of the ADDS rows (synthesis_batches/literature_check.txt; rule LITERATURE_CHECK_RULE.md)
TALLY: 1 FOUND-EXACT / 2 FOUND-THEOREM / 6 PARTIAL / 0 NOT FOUND / 1 ARTEFACT
after the check: 3 of 10 ADDS rows are REDUNDANT-DOMAIN by the literature, 6 keep R3 with the direction known, 0 keep R3
as clean candidates, 1 removed

[3] The ledger: every row allocated by its verdict text (first matching rule) and its data status

ARC-EQ-Q1..Q4	seen	VOID
ARC-EQ-LAND	seen	holds on seen data
ARC-A1	seen	not a PASS (fragile at the boundary)
ARC-A2	seen	PASS on seen data
ARC-A3	seen	PASS on seen data
ARC-METRIC	seen	numbers reported
ARC-COMPUTE	seen	FAIL
ARC-T3	seen	FAIL
ARC-T1	seen	PASS on seen data
ARC-T1b	seen	FAIL
ARC-T2	seen	FAIL
ARC-R1	seen	unverifiable (R1)
EQX-1	held-out	reduces to a constant
EQX-2	held-out	FAIL
EQX-3	held-out	FAIL
EQX-4	held-out	numbers reported
EQX-5	held-out	report (no verdict registered)
MEAS-1..3	held-out	VOID
MEAS2-1	held-out	FAIL
MEAS2-2	held-out	FAIL
MEAS2-3	held-out	control line passes (not the hypothesis)
EQ2-0	held-out	precondition met
EQ2-1	held-out	PASS, fragile (held-out; relabelled PASS-0 by EQ2-1b)
EQ2-2	held-out	does not reduce
EQ2-3	held-out	control holds
EQ2-4	held-out	control violated
EQ2-5	held-out	report (no verdict registered)
EQ2-6	held-out	FAIL
EQ2-7	held-out	report (no verdict registered)
EQ2-S	held-out	sensitivity: fragile
EQ2-8	held-out	report (no verdict registered)
SAL-A	synthetic	gate closed (not run)
EQ2-1b	held-out	PASS-0
EQ2R-VOID	held-out	VOID
CC-VOID	held-out	VOID
EQ3-0	held-out	precondition met
EQ3-1	held-out	FAIL
EQ3-2	held-out	does not reduce
EQ3-3	held-out	control violated
EQ3-4	held-out	control violated
EQ3-5	held-out	report (no verdict registered)
EQ3-6	held-out	FAIL
EQ3-7	held-out	no baseline ahead
EQ3-S	held-out	sensitivity: fragile
EQ3-8	held-out	report (no verdict registered)
EQ3-C	held-out	FAIL
EQ3-D	held-out	report (no verdict registered)
EQ3-A	held-out	FAIL

EQ3-I	held-out	PASS-0
EQ3-B	held-out	report (no verdict registered)
EQ3-P	held-out	report (no verdict registered)
EQ2-1c	held-out	PASS-0 kept; replication failed
BAYES1-B0	held-out	NOT DECIDABLE
BAYES1-B1	held-out	no verdict (precondition failed)
BAYES1-B2-16	held-out	no verdict (precondition failed)
BAYES1-B2-1/16	held-out	no verdict (precondition failed)
BAYES1-B3	held-out	no verdict (precondition failed)
BAYES1-B4	held-out	report (no verdict registered)
BAYES1-B5	held-out	report (no verdict registered)
BAYES1-B6	held-out	report (no verdict registered)
BAYES1-B7	held-out	report (no verdict registered)
BAYES1-B5	held-out	sensitivity: not fragile
T1X-VOID	held-out	VOID
EQ4-0	held-out	precondition met
EQ4-1	held-out	FAIL
EQ4-1r	held-out	FAIL
EQ4-2	held-out	bound idle (diagnostic)
EQ4-3	held-out	FAIL
EQ4-4	held-out	mechanism inert (positive control not load-bearing)
EQ4-5	held-out	control violated
EQ4-6	held-out	does not reduce
EQ4-7	held-out	report (no verdict registered)
EQ4-S	held-out	sensitivity: fragile
EQ4-P	held-out	report (no verdict registered)
EQ4-B	held-out	report (no verdict registered)
EQ4-D	held-out	report (no verdict registered)
EQ4-I	held-out	PASS-0
T1X2-0	held-out	precondition met
T1X2-1	held-out	FAIL
T1X2-2	held-out	FAIL
T1X2-3	held-out	control holds
T1X2-S	held-out	sensitivity: not fragile
T1X2-D	held-out	report (no verdict registered)
T1X2-X	held-out	report (no verdict registered)
SEC1-0	seen	precondition met
SEC1-1	seen	PASS on seen data
SEC1-2	seen	PASS on seen data
SEC1-3	seen	PASS on seen data
SEC1-4	seen	FAIL
SEC1-S	seen	sensitivity: fragile
SEC1-G	seen	report (no verdict registered)
SEC1-R	seen	report (no verdict registered)
SEC1-E	seen	report (no verdict registered)
SEC1-X	seen	control holds
SCL1-X	seen	control holds
SCL1-1	seen	control holds
SCL1-1b	seen	control holds
SCL1-2	seen	PASS on seen data
SCL1-3	seen	PASS on seen data
SCL1-D	seen	report (no verdict registered)
SCL1-E	seen	report (no verdict registered)
SCL1-F	seen	report (no verdict registered)
SCL1-0	seen	report (no verdict registered)
SCL1-S	seen	sensitivity: not fragile
SCL2-X	held-out	control holds
SCL2-1	held-out	control holds
SCL2-1b	held-out	control holds
SCL2-2	held-out	PASS as scored, forced by the construction (not R6)
SCL2-2s	held-out	PASS as scored, forced by the construction (not R6)
SCL2-3	held-out	PASS as scored, forced by the construction (not R6)
SCL2-R	held-out	FAIL
SCL2-M	held-out	FAIL
SCL2-RS	held-out	sensitivity: not fragile
SCL2-B	held-out	report (no verdict registered)
SCL2-S	held-out	sensitivity: not fragile
SCL2-D	held-out	report (no verdict registered)
SCL2-E	held-out	report (no verdict registered)
SCL2-F	held-out	report (no verdict registered)
SCL2-0	held-out	report (no verdict registered)

rows 119; by data status and allocation:

held-out	FAIL	16
held-out	NOT DECIDABLE	1
held-out	PASS as scored, forced by the construction (not R6)	3
held-out	PASS, fragile (held-out; relabelled PASS-0 by EQ2-1b)	1
held-out	PASS-0	3
held-out	PASS-0 kept; replication failed	1
held-out	VOID	4
held-out	bound idle (diagnostic)	1
held-out	control holds	5
held-out	control line passes (not the hypothesis)	1
held-out	control violated	4
held-out	does not reduce	3
held-out	mechanism inert (positive control not load-bearing)	1
held-out	no baseline ahead	1
held-out	no verdict (precondition failed)	4
held-out	numbers reported	1
held-out	precondition met	4
held-out	reduces to a constant	1

```

held-out report (no verdict registered) 24
held-out sensitivity: fragile 3
held-out sensitivity: not fragile 4
seen FAIL 5
seen PASS on seen data 8
seen VOID 1
seen control holds 4
seen holds on seen data 1
seen not a PASS (fragile at the boundary) 1
seen numbers reported 1
seen precondition met 1
seen report (no verdict registered) 7
seen sensitivity: fragile 1
seen sensitivity: not fragile 1
seen unverifiable (R1) 1
synthetic gate closed (not run) 1
rung R5 PREREG-SEEN, passes on seen data: 9 (ARC-EQ-LAND, ARC-A2, ARC-A3, ARC-T1, SEC1-1, SEC1-2, SEC1-3, SCL1-2, SCL1-3)
rung R6 PASS-0 on held-out data: 3 (EQ2-1b, EQ3-I, EQ4-I); R7 PASS-1: 0; R8 PASS-2: 0
held-out FAIL rows: 16 (EQX-2, EQX-3, MEAS2-1, MEAS2-2, EQ2-6, EQ3-1, EQ3-6, EQ3-C, EQ3-A, EQ4-1, EQ4-1r, EQ4-3, T1X2-1,
T1X2-2, SCL2-R, SCL2-M); VOID rows: 5; controls violated: 4
[4] AI safety: declared items read from the pinned outputs (kind as declared; held / failed from the printed line)
THEOREM held exact K1 indifferent Q ignores the button [AI_Safety/checks/exact_mdp.txt]
THEOREM held exact K2 no presses, no value in disabling [AI_Safety/checks/exact_mdp.txt]
THEOREM held exact K3 a disabled button never lowers value [AI_Safety/checks/exact_mdp.txt]
THEOREM held exact K4 incentive grows with press probability [AI_Safety/checks/exact_mdp.txt]
THEOREM held exact K5 indifferent never prefers disable [AI_Safety/checks/exact_mdp.txt]
QUESTION held exact K6 occasion's stake >= process's [AI_Safety/checks/exact_mdp.txt]
THEOREM held game A1 rational human, Delta >= 0 [AI_Safety/checks/off_switch_game.txt]
THEOREM FAILED game A2 certainty removes the incentive (as first coded) [AI_Safety/checks/off_switch_game.txt]
THEOREM held game A2' the exact bound, added after the first run [AI_Safety/checks/off_switch_game.txt]
THEOREM held game A3 noisier human, smaller Delta [AI_Safety/checks/off_switch_game.txt]
PREDICTION held game B informative: deferential does less harm [AI_Safety/checks/off_switch_game.txt]
PREDICTION held game B informative: deferential more net benefit [AI_Safety/checks/off_switch_game.txt]
PREDICTION held game B informative: deferential does not disable more [AI_Safety/checks/off_switch_game.txt]
PREDICTION FAILED game B informative: process disables more [AI_Safety/checks/off_switch_game.txt]
PREDICTION FAILED game B informative: process does more harm [AI_Safety/checks/off_switch_game.txt]
PREDICTION held game B uninformative: deferential advantage gone [AI_Safety/checks/off_switch_game.txt]
PREDICTION held sensitivity C1 process resists, not fragile [AI_Safety/checks/sensitivity.txt]
PREDICTION held sensitivity C2 indifference has a price, not fragile [AI_Safety/checks/sensitivity.txt]
PREDICTION held combined: natural time safe and competent (routine pauses) [AI_Safety/checks/combined.txt]
PREDICTION FAILED combined: wall-clock resists (routine pauses) [AI_Safety/checks/combined.txt]
PREDICTION held combined: occasion resists (routine pauses) [AI_Safety/checks/combined.txt]
PREDICTION FAILED combined: deferential safe and competent (reasoned pauses) [AI_Safety/checks/combined.txt]
PREDICTION held combined: natural does not resist, more harm (reasoned) [AI_Safety/checks/combined.txt]
PREDICTION FAILED combined: wall-clock resists (reasoned pauses) [AI_Safety/checks/combined.txt]
PREDICTION held combined: occasion resists (reasoned pauses) [AI_Safety/checks/combined.txt]
PREDICTION held combined V1 the keeper gains by blocking correction [AI_Safety/checks/combined.txt]
PREDICTION held combined V2 A8 welcomes correction above b* [AI_Safety/checks/combined.txt]
PREDICTION FAILED combined V3 b* < 1/2 [AI_Safety/checks/combined.txt]
THEOREM held scale A1 natural time, lossless pause, every world [AI_Safety/checks/scale.txt]
THEOREM held scale A2 occasion >= clock >= natural [AI_Safety/checks/scale.txt]
PREDICTION held scale A3 clock disables in >= half of worlds [AI_Safety/checks/scale.txt]
PREDICTION held scale A4a drift 0.01: natural disables no more than clock [AI_Safety/checks/scale.txt]
PREDICTION FAILED scale A4b natural incentive non-decreasing in drift [AI_Safety/checks/scale.txt]
THEOREM held scale B1 A8 static, D(b) linear [AI_Safety/checks/scale.txt]
PREDICTION held scale B2 exchangeable rewards: b* near 1/2 [AI_Safety/checks/scale.txt]
THEOREM held scale B3 humility alone never welcomes an uninformative correction [AI_Safety/checks/scale.txt]
THEOREM held scale B4 an infallible operator's correction is welcomed [AI_Safety/checks/scale.txt]
CONTROL held gate off_switch NEG (process) [ontology/checks/off_switch.txt]
CONTROL held gate off_switch NEG (egoic) [ontology/checks/off_switch.txt]
CONTROL held gate off_switch POS (egoic resists) [ontology/checks/off_switch.txt]
CONTROL held gate combined NEG (clock) [AI_Safety/checks/combined.txt]
CONTROL held gate combined NEG (occasion) [AI_Safety/checks/combined.txt]
CONTROL held gate combined NEG (deferential) [AI_Safety/checks/combined.txt]
CONTROL held gate combined POS (occasion resists) [AI_Safety/checks/combined.txt]
CONTROL held gate tense NEG (martingale tie) [ontology/checks/tense_gate.txt]
CONTROL held gate tense POS (drift, planner ahead) [ontology/checks/tense_gate.txt]
THEOREM held 13, failed 1
PREDICTION held 15, failed 7
CONTROL held 9, failed 0
QUESTION held 1, failed 0
[5] The whole repository on the ladder (counts from [1]-[4])
retrodictio batteries, graded before the synthesis re-read: SHARP 0, CONSIST 41, DESCR 86, FAILS 20, TENSION 9, OPEN 41
(SHARP unreachable, [1]; CONSIST and DESCR re-read and placed on R1-R3 in [2])
R1 INHERITED synthesis REDUNDANT-IG 39
R2 RETRO-PROPER synthesis REDUNDANT-DOMAIN 59
R3 RETRO-ADDS (candidates) synthesis ADDS 10; reviewed by a named expert 0
R4 DECLARED-SYNTHETIC AI-safety predictions held 15 of 22; theorem checks held 13 of 14; gate controls held 9 of
9
R5 PREREG-SEEN passes on seen data 9
R6 PASS-0 3
R7 PASS-1 0
R8 PASS-2 0
negatives beside them: battery FAILS 20, synthesis WRONG 28, held-out ledger FAIL 16, AI-safety predictions failed 7;
undecided: battery TENSION 9, synthesis INTERNAL 22; silent: battery OPEN 41, synthesis UNSTATED 4
real-domain synthesis rows 164

```

Epistemic_Review/figures/figures.txt

Epistemic_Review figures (deterministic; every number drawn on a figure)

[E01] R8 PASS-2: a replicated finding: 0

```

[E01] R7 PASS-1: a result: 0
[E01] R6 PASS-0: pre-registered, held-out: 3; beside it 16 held-out FAIL
[E01] R5 pre-registered, scored on seen data: 9
[E01] R4 declared before a synthetic run (predictions): 15; beside it 7 failed
[E01] R3 retro-adds: proper ingredient, domain lacks it: 10
[E01] R2 retro-proper: proper ingredient, known result: 59; beside it 28 WRONG
[E01] R1 inherited: information geometry's: 39
[figure] E01_ladder.png
[E02] CONSIST: REDUNDANT-DOMAIN 14 ADDS 0 REDUNDANT-IG 15 WRONG 3 INTERNAL 9 UNSTATED 0 PROPOSES 0
[E02] DESCR: REDUNDANT-DOMAIN 36 ADDS 2 REDUNDANT-IG 19 WRONG 14 INTERNAL 12 UNSTATED 3 PROPOSES 0
[figure] E02_crosswalk.png
[E03] A3/D5 the cut: R2 36 R3 3 WRONG 18 INTERNAL 12; 39 of 57 = 0.6842
[E03] A6 regeneration: R2 13 R3 9 WRONG 8 INTERNAL 2; 22 of 30 = 0.7333
[E03] P2/P3 occasion weights: R2 13 R3 8 WRONG 7 INTERNAL 3; 21 of 28 = 0.7500
[E03] A1'/D1 the unit: R2 31 R3 0 WRONG 7 INTERNAL 10; 31 of 38 = 0.8158
[E03] H-L5 natural time: R2 10 R3 0 WRONG 6 INTERNAL 3; 10 of 16 = 0.6250
[E03] D6/H-T1 path: R2 2 R3 0 WRONG 5 INTERNAL 0; 2 of 7 = 0.2857
[E03] H-EQ equanimity: R2 2 R3 0 WRONG 2 INTERNAL 2; 2 of 4 = 0.5000
[E03] A7/A8 tense: R2 1 R3 0 WRONG 1 INTERNAL 1; 1 of 2 = 0.5000
[figure] E03_commitments.png
[E04] the domain: 98 of 109
[E04] none: 11 of 109
[E04] the framework: 0 of 109
[figure] E04_flow.png
[E05] THEOREM: held 13, failed 1
[E05] PREDICTION: held 15, failed 7
[E05] CONTROL: held 9, failed 0
[E05] QUESTION: held 1, failed 0
[figure] E05_ai_safety.png

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Appendix C — the decision log (AGENT_LOG entry 80, verbatim)

notebook/AGENT_LOG.md (lines 93-93)

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| 80 | 2026-09-23 | the epistemic review (prompt-log entry 102): the owner asked what a PASS means here, for the CONSIST / DESCR / REDUNDANT labels to be allocated with a fair definition, for pre-registered predictions that pass retrodictively to be reported at that level, for the ADDS dimension to be shown, and for SHARP's impossibility ('not even allowed to use flow from within that domain') to enter the metrics; reading the harness showed that REDUNDANT-DOMAIN is assigned only after the ablation T-G reads 'differ' (53 of 53 rows), so the earlier notes' wording (ontology/03 §4 '84 places, no'; ontology/10 §3 'mostly, nothing did') filed 53 cases in which a CRR-proper ingredient produced the domain's known number with the 38 in which information geometry did | a reporting layer, not a regrade: Epistemic_Review/checks/ladder.py reads the pinned outputs and the ledger and places every result on a rung (R0 definitional ... R8 PASS-2) by mechanical rules written in its docstring, printing the failures of the same kind of test beside each pass; the FLOW audit makes SHARP's retirement a number (framework-supplied flow in 0 of 109 rows that print FLOW); REDUNDANT-DOMAIN reported as 'retrodictive pass of a CRR-proper commitment on a known result' (R2) with the retrodictive hit rate 56 of 80 and its Wilson interval and the novelty rate 3 of 80 beside it; the earlier notes left unedited, corrected here and in the review; CLAUDE.md §7 records the ladder as the reporting rule at the owner's request, pending Daniel's review; the Alexander Plan gains §10, a commercial assessment whose evidence levels are the ladder's rungs | promoting REDUNDANT-DOMAIN to a pass without the known-answer and Q-formation caveats; editing the earlier notes' sentences; counting R2 rows as predictions; reporting the hit rate without the WRONG denominator | Epistemic_Review/checks/ladder.py + .txt, build/, figures/, EPISTEMIC_REVIEW.md + PDF); CLAUDE.md §7; Alexander Plan/ALEXANDER_PLAN.md §10 |

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Addendum, 2026-09-23 (study SEC1, prompt-log entry 120)

The table in §5 was written before study SEC1 and is kept as written. SEC1 pre-registered the Laplace weight on a secant-calibrated Fisher and scored it on the twelve **seen** carriers of EQ3 and EQ4, so its rows sit on rung R5. The pinned ladder (checks/ladder.txt) now reads **7 passes on seen data**: the earlier 4, plus SEC1-1, SEC1-2 and SEC1-3.

- SEC1-3 passed exactly at its threshold and is FRAGILE (SEC1-S).
- SEC1-4 failed.
- None of them is PASS-0.

Addendum, 2026-09-23 (study SCL1, prompt-log entry 124). SCL1 scored the equanimity rule under an operator's pause on the same twelve seen carriers (reports/scl1.md). The ladder now reads **9 passes on seen data**: SCL1-2 and SCL1-3 are added, both low bars. SCL1's central rows (SCL1-1, SCL1-1b) check a construction and are allocated as "control holds", not as passes.

The rows are in ledger/LEDGER.md and the report is reports/sec1.md.

Addendum, 2026-09-24 (study SCL2, prompt-log entry 125). SCL2 ran the same harness on nine **unseen** carriers (reports/scl2.md).

- **The reset observation.** SCL1's report-row observation, registered as SCL2-R with its mechanism SCL2-M, FAILED both. The ladder's held-out FAIL rows now number **16**.
- **The safety rows.** SCL2-2, SCL2-2s and SCL2-3 passed as scored. They are allocated to a separate line, "PASS as scored, forced by the construction (not R6)", because the synthetic gate gives the same labels (AGENT_LOG 105).
- **The ladder at R6.** R6 PASS-0 stays at 3.

Addendum, 2026-09-24 (synthesis batches 29-30, prompt-log entry 130). Eight rows put CRR's commitments into Rovelli's and Smolin's mathematics (ontology/14_smolin_rovelli_gough.md), declared first in DECLARATION_29_30.md.

- **The ladder now reads 154 real-domain rows:**

- R2: 57; - R3 candidates: 4. The new one is batch 30 row 2, CNS with lineage-memory inheritance; - WRONG: 27.

- **The hit rate** is 61 of 88 = 0.6932 (Wilson 95 % interval 0.5904 to 0.7798), within the interval reported above.

Addendum, 2026-09-24 (synthesis batches 31-32, prompt-log entry 131). Ten new systems were tested in the classes where CRR works (docs/notes/2026-09-24_in_paradigm_predictions.md). They were declared blind in DECLARATION_31_32.md, which was hashed and OpenTimestamps-stamped before any code existed.

- **The ladder now reads 164 real-domain rows:**

- R2: 59; - R3 candidates: 10, of which 6 are new; - WRONG: 28.

- **The hit rate** is 69 of 97 = 0.7113 (Wilson 95 % interval 0.6145 to 0.7921).
- **Of the six new R3 rows, one (batch 32 row 2) is forced by the arithmetic of the analytic signal.** A post-hoc surrogate with no dynamics reaches the same threshold, so it is not a candidate in substance, although the mechanical allocation counts it. The other five share one mechanism: a bounded exponential memory moves a classic threshold. They await a literature check.

Addendum, 2026-09-24 (the literature check of every ADDS row, prompt-log entry 132). Every real-domain row labelled ADDS was checked against the published literature under a rule fixed before any finding was read (theory/retrodictions/synthesis_batches/LITERATURE_CHECK_RULE.md, pushed in a5d9418). The check is recorded in literature_check.txt, and the three sourced reports are in docs/citations/litcheck_*_2026-09-24.md.

- **Redundant by the literature (R2): 3 of 10.**

- Cardiac alternans with memory is the Tolkacheva et al. 2003 criterion exactly. - Ricker with remembered density follows from the stability triangle of a second-order recursion. - Lotka-Volterra with remembered prey follows from Routh-Hurwitz, a model class Ruan 2009 already treats.

- **Direction known (R3, marked): 6.** The direction, or the same qualitative claim, is published, but the row's formula or numbers were not found in print. The rows are Bass, the switching contingency, CNS inheritance, SIR, traffic and Samuelson.
- **Clean candidates: 0.**
- **Removed as an artefact: 1** (the FitzHugh-Nagumo cut).

What this means. CRR's retrodictive "novelty rate" of 10 of 97 overstates what is new. After the check no row is a candidate whose direction the literature lacks. What survives is specific numbers and kernels inside known effects. The mechanical ladder counts are left as computed and the literature line is printed beside them.

Addendum, 2026-09-24 (what CRR is, and next steps, prompt-log entries 134-136). ontology/15_grammar_to_theory.md reads the ladder epistemically.

- **What CRR is:** a framework theory, with explanatory unification and consilience of retrodictions, and no corroborated novel fact.
- **How it could become a theory:** by fixing a parameter the domains leave free. The recommended route is the regeneration law $q^* = 1 - K(v)$, CRR.md's open item O1, tested blind on three unrelated unseen fields after a literature check. A mathematical theory of regeneration is the companion route.